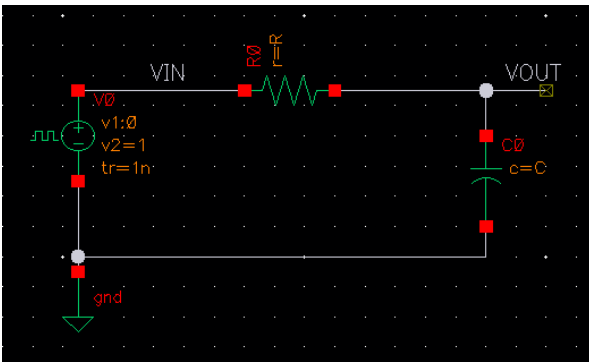


Content

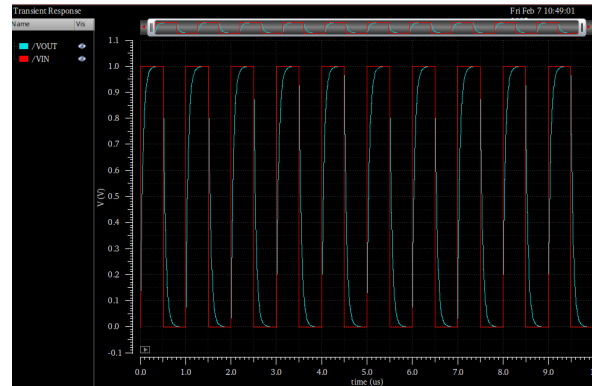
- Cadence introduction
 - Connecting to server
 - Create a Voltage Divider
 - Run DC simulations
 - Run Transient simulation on RC-filter
 - Run AC simulations
 - Plotting waveforms
 - Performing FFT
 - Parametric analysis
 - Reference
- Exercise 1
- Exercise 2
- Exercise 3
- Exercise 4
- Exercise 5

What is Cadence?

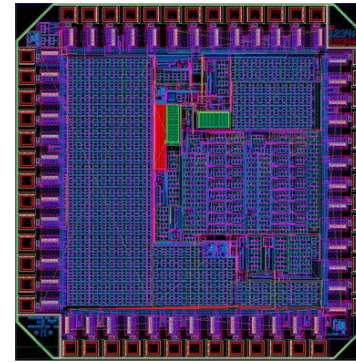
- Cadence is a circuit design software widely used in the industry.
- It can be used for schematic design, schematic-level circuit simulation, layout design, and post-layout simulation, all the way to complete chip design.
- In this course, we will use Cadence for schematic-level simulation



Schematic design



Circuit simulation

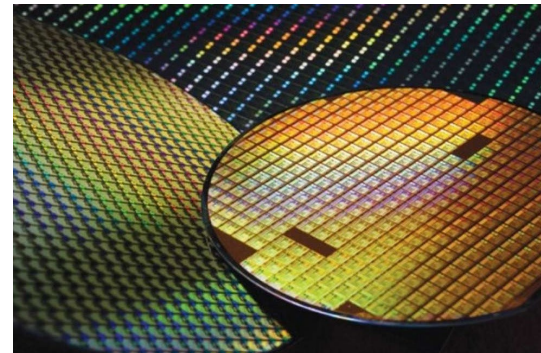


Layout design

Post-layout
simulation

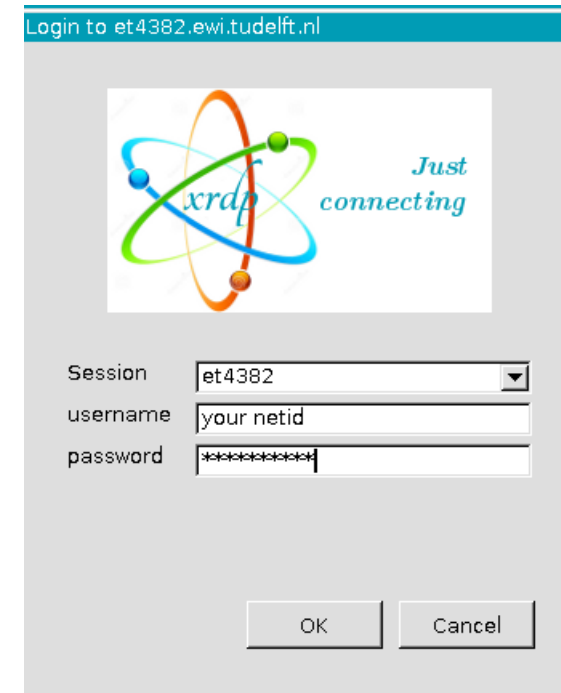
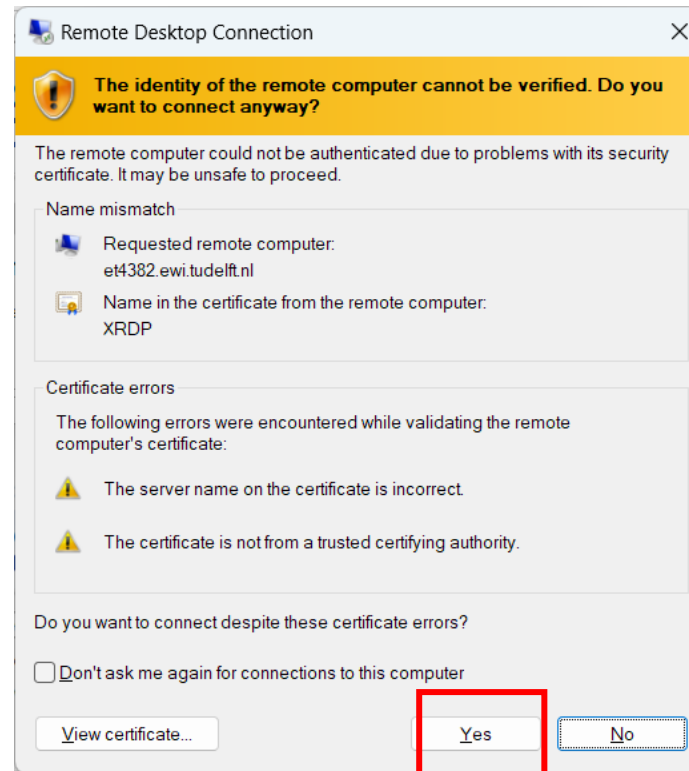
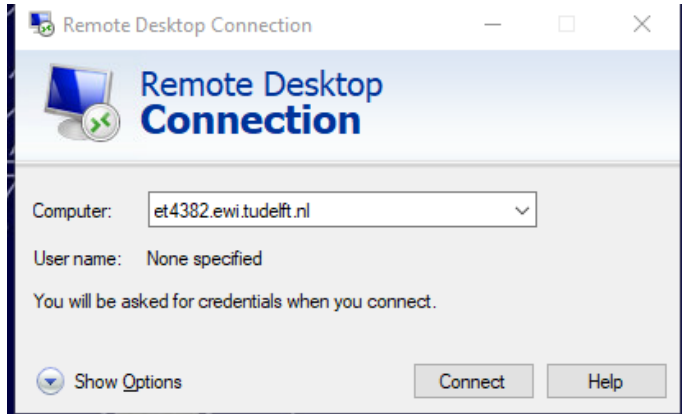
We are using a new Cadence version now, any mentioning of “ADE L” refers to “ADE Explorer” now

Tapeout



Connect to server

- Use remote desktop connection and type in “et4382.ewi.tudelft.nl”. Then click connect
- Type in you netid and netid password



Set your environment

First, set up your working directory.

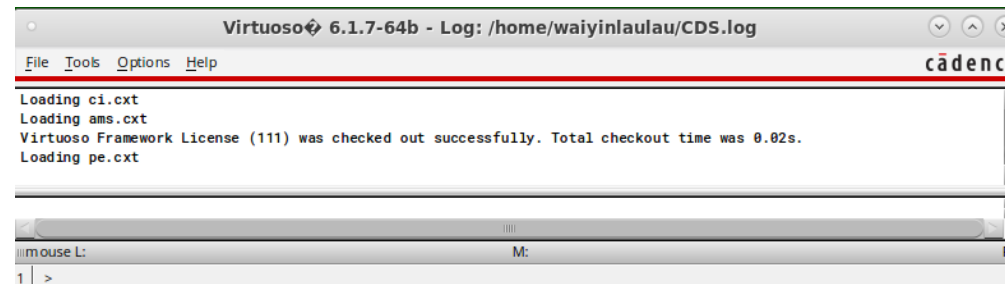
You only need to do this once. (Hit ENTER after each line.)

```
cd ~  
mkdir tsmcBCD  
cd ~/tsmcBCD  
source /opt/ei/DK/tsmc/muse/BCDG2/2.16B_5M6_40K/et4382/start
```

To start Cadence, type the following.

```
cd ~/tsmcBCD  
source sourceme  
virtuoso&
```

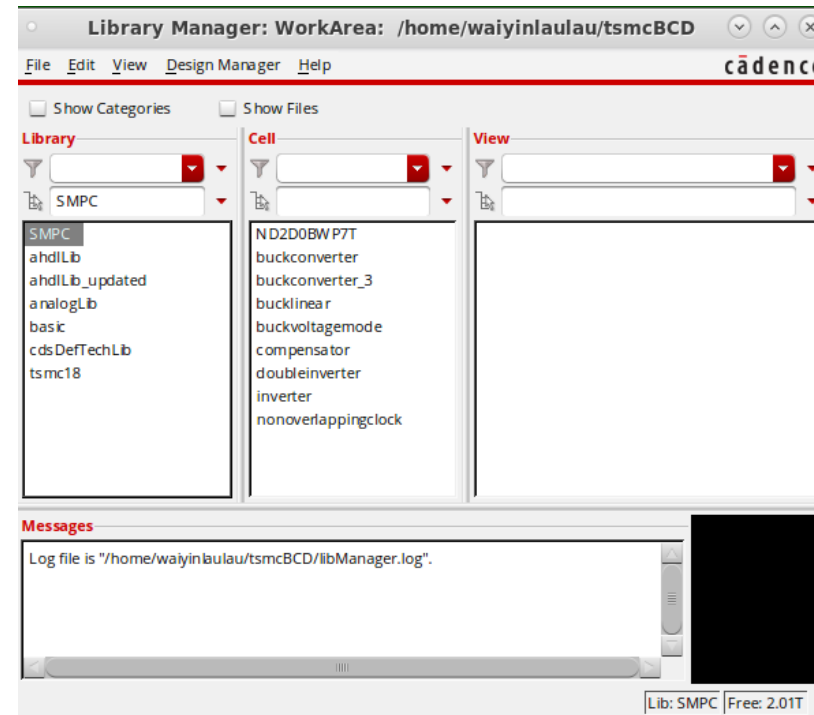
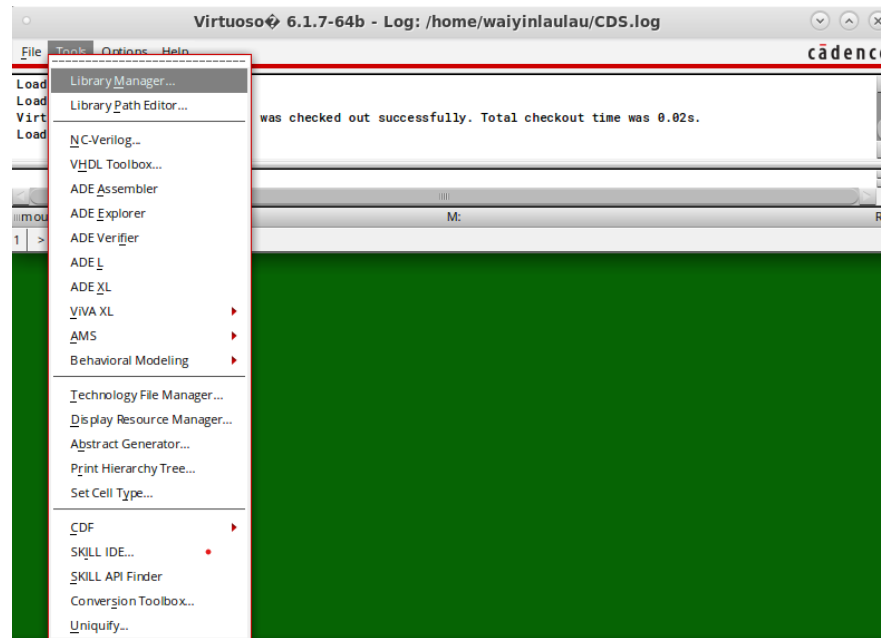
You need to do this every time you start Cadence. After Cadence starts, you should see the CIW window:



Tips: For Linux terminal, You can use Key "Up Arrow" to show last command entered in your terminal history, Vice verse with Key "Down Arrow". The key is located on your keyboard. This can save your time if you need to reuse some commands.

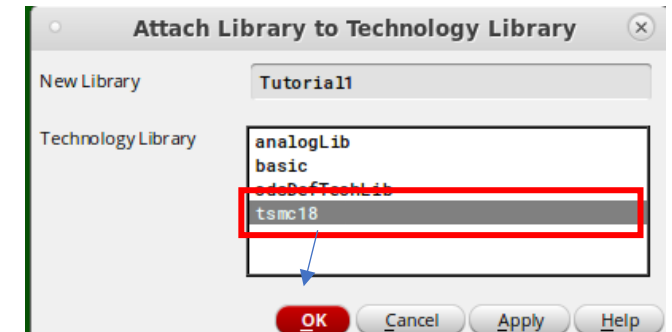
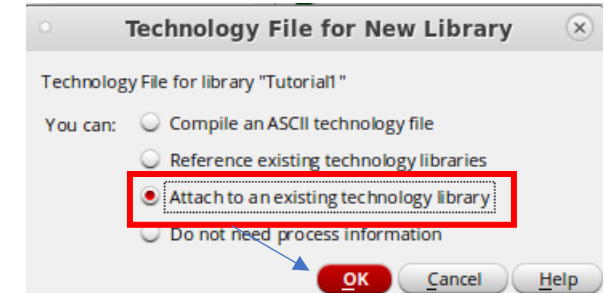
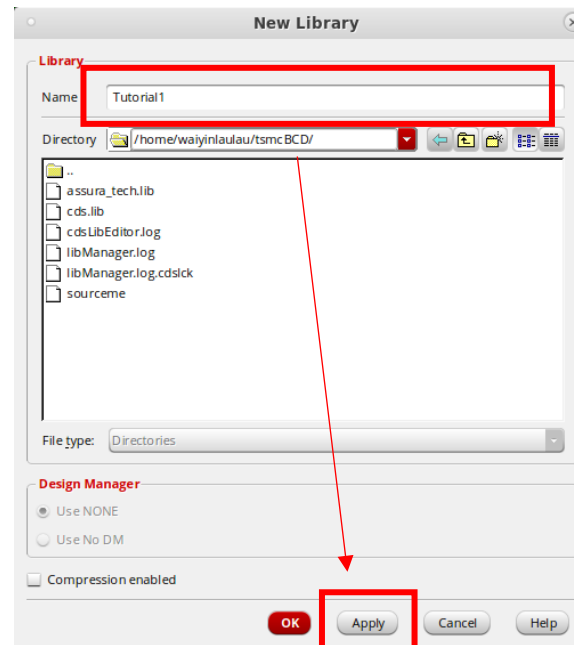
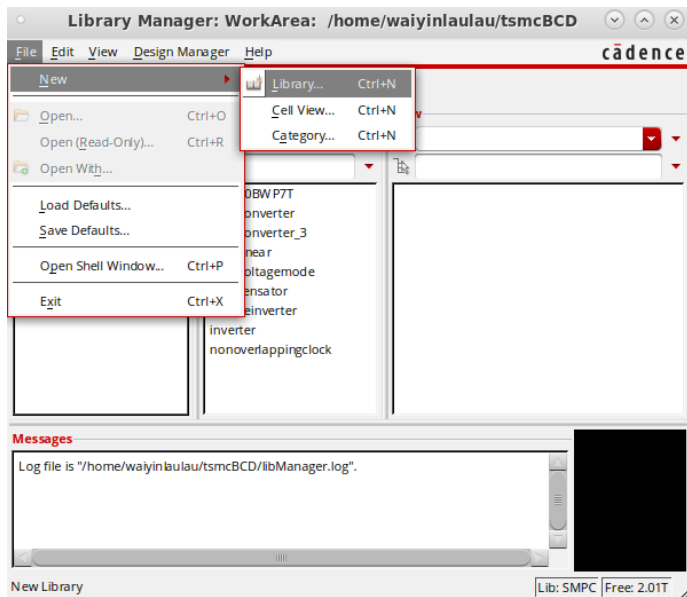
Library Manager

- In CIW, click "Tool" → "Library Manager" to open library manager, where it shows all available libraries to use
- For example, in library "analogLib", it stores ideal analog instants
- For convenience and proper management of components, we will create a library for this tutorial1.



Creation of library "Tutorial1"

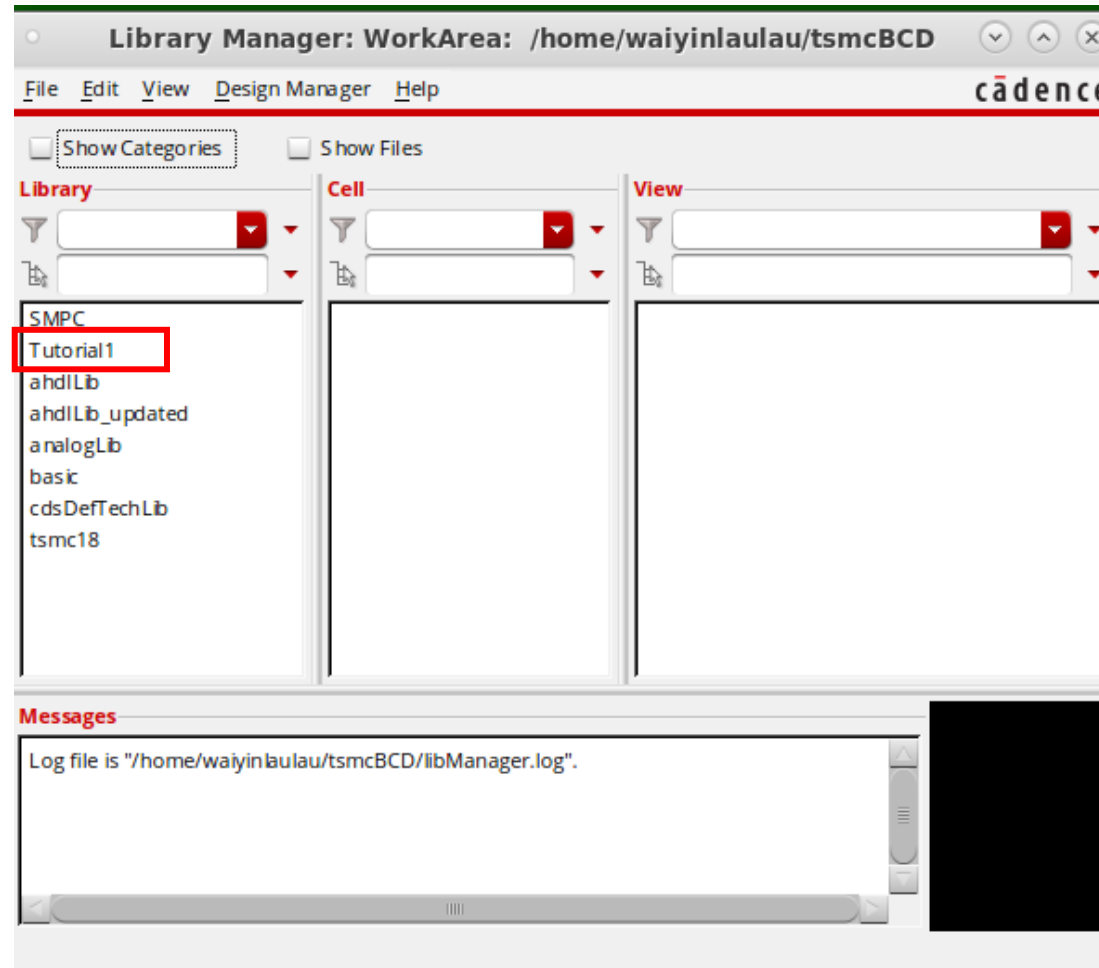
- Attaching to "tsmc18" means everything in this "Tutorial1" library is restricted under the rules of TSMC180nm PDK, Which you signed NDA for. For example, limitation on MOSFET width or length



If Window pops up for "TSMC PDK setting information", just click **close** to close it.

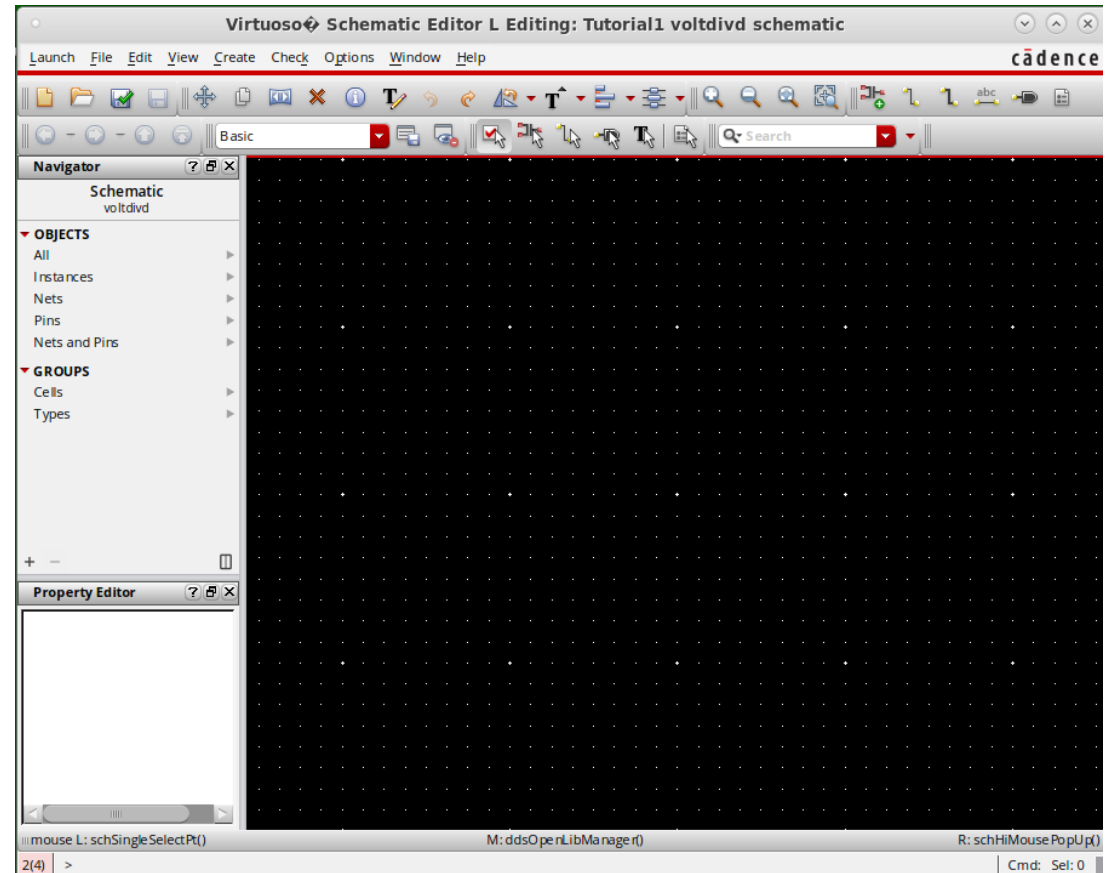
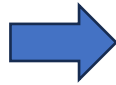
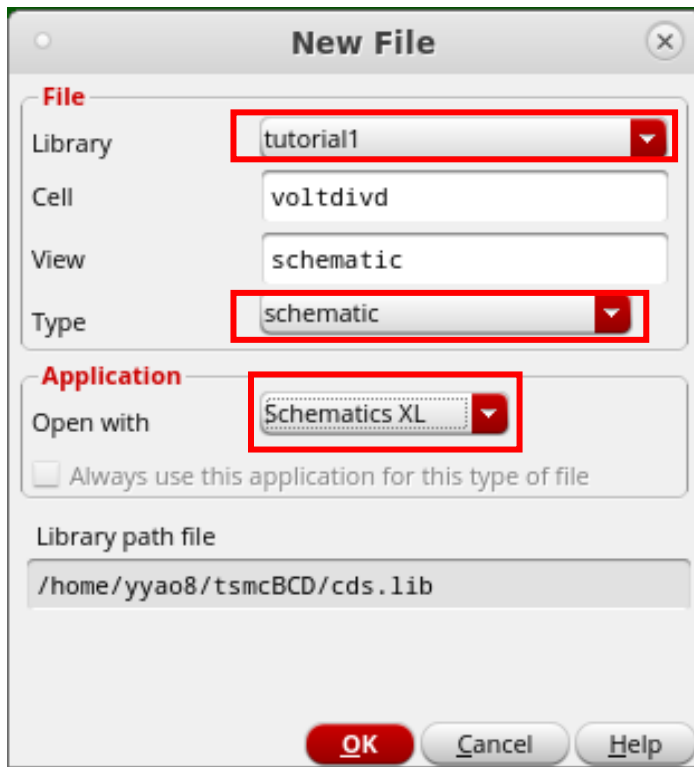
Successful creation of library

- Now we should see a library called "Tutorial1" in Library Manager
- Next, we need to create a schematic



Schematic creation

- Schematic is where we place our electrical components
- In Library manager, click on **File** → **New** → **Cellview**



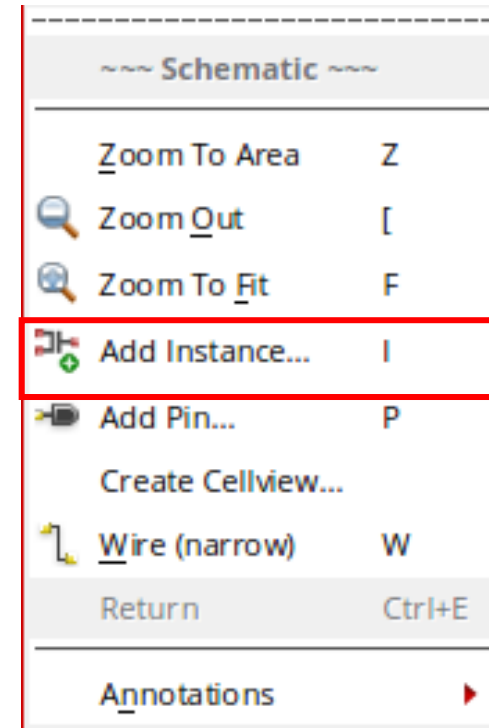
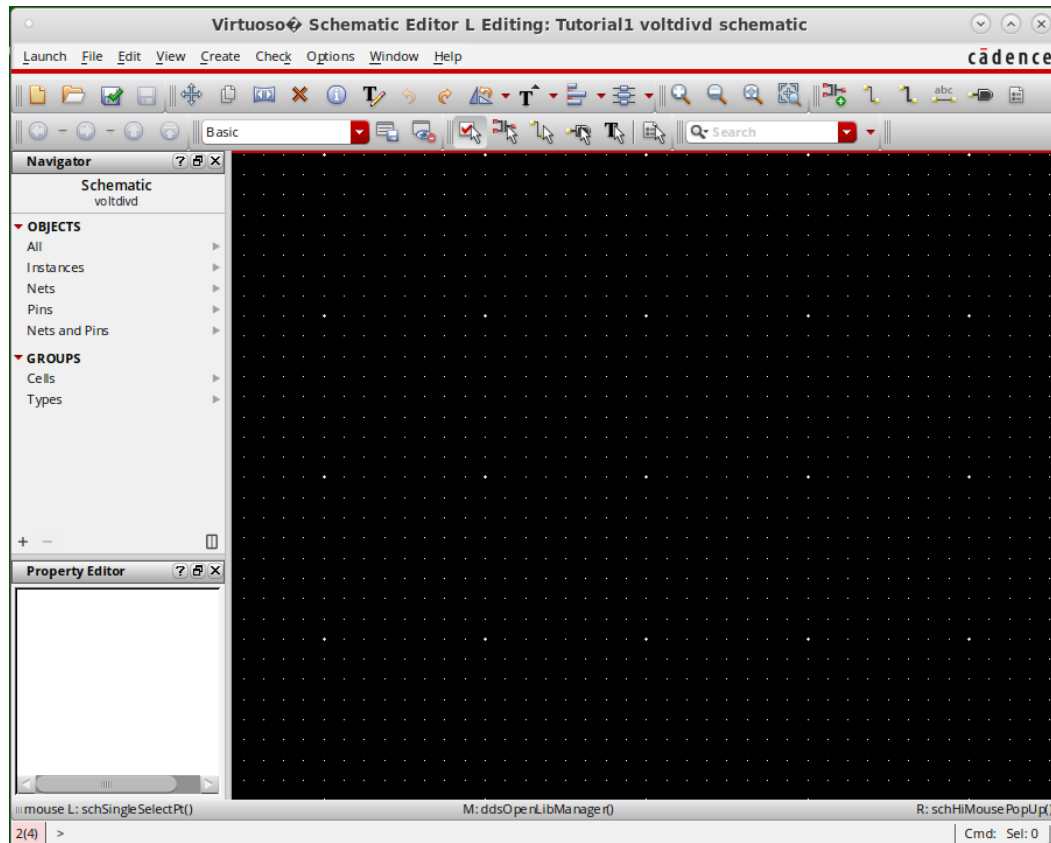
Exercise1:Voltage divider

Action Item:

- Server login
- Create a library
- Create a schematic and insert instance for a voltage divider

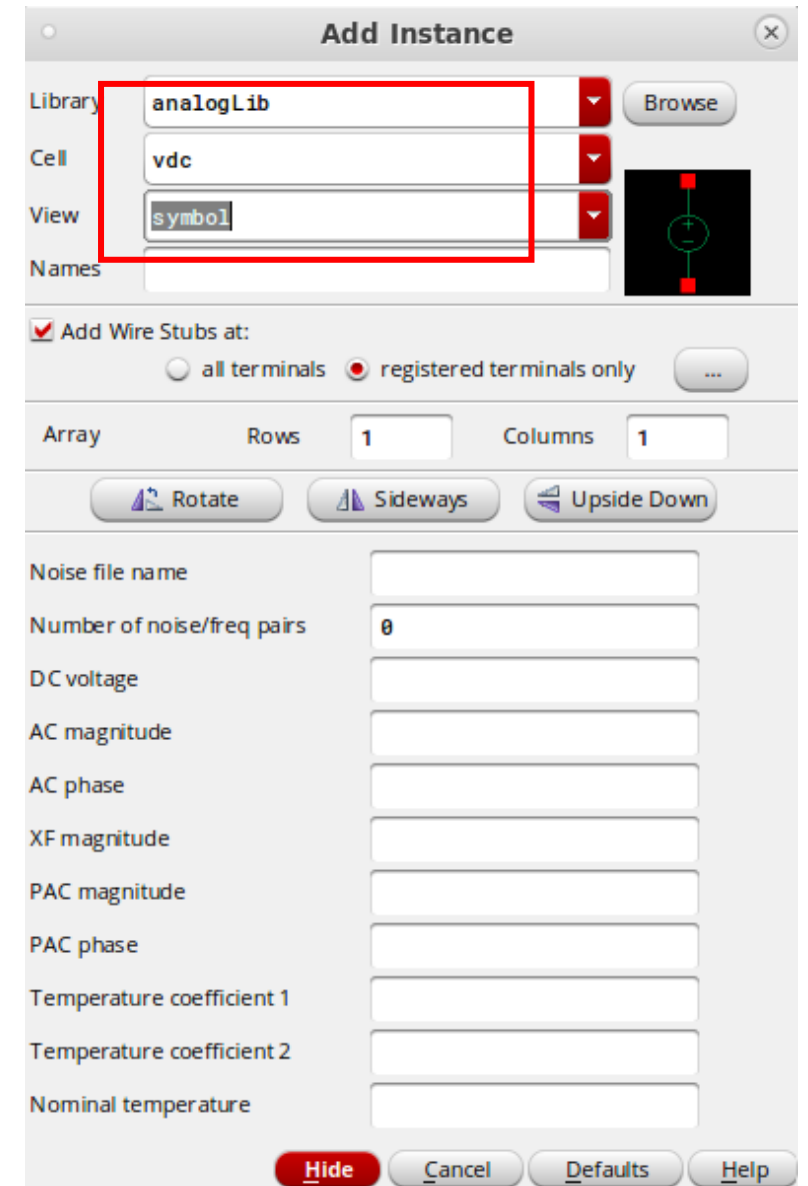
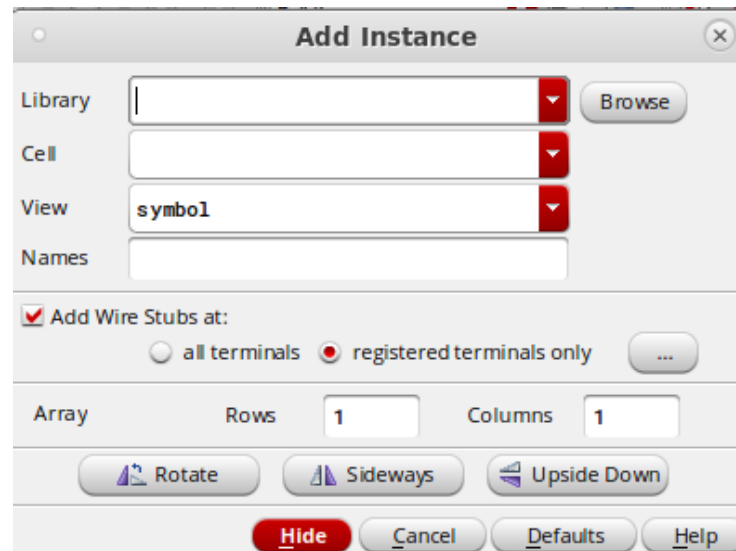
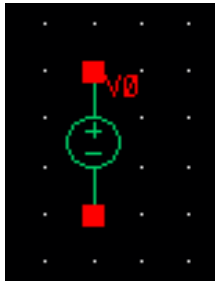
Instance insertion

- Add instance:
 - Right click, a pull-down menu show, then click "Add Instance"
 - Or Hotkey "i"



Instance insertion: DC source

- An “Add instance” window will pop-up
- Search for ideal voltage source: analogLib - vdc
- Choose View “symbol ” to add this voltage source into schematic



Instance modification

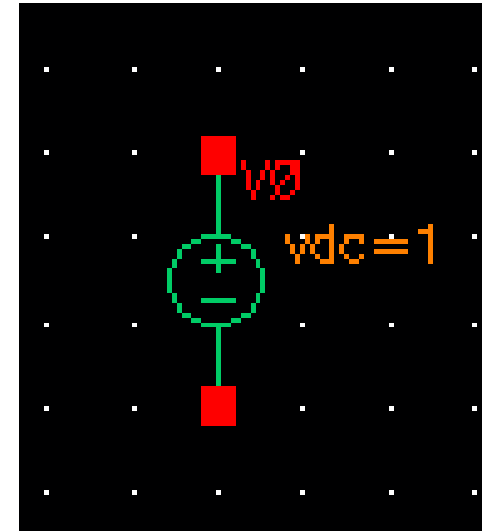
- To modify instance: hit hotkey "q"
- A window called **Edit object properties** will pop up. Modify the value of **DC voltage** to "1"
- Click OK, the voltage source will show "vdc=1"

The 'Edit Object Properties' dialog box is shown. It has a title bar with a close button. The 'Apply To' section has 'only current' and 'instance' dropdowns. The 'Show' section has checkboxes for 'system', 'user', and 'CDF'. Below are 'Browse' and 'Reset Instance Labels Display' buttons. A table lists properties: Library Name (analogLib), Cell Name (vdc), View Name (symbol), and Instance Name (V0), each with a 'Display' dropdown set to 'off'. Below this is a section for 'User Property' with 'Add', 'Delete', and 'Modify' buttons, and a table with 'Ivignore' set to 'TRUE'. The 'CDF Parameter' section has a table with 'DC voltage' set to '1', which is highlighted with a red box. At the bottom are 'OK', 'Cancel', 'Apply', 'Defaults', 'Previous', 'Next', and 'Help' buttons. A blue arrow points from the 'OK' button to the 'DC voltage' field.

Property	Value	Display
Library Name	analogLib	off
Cell Name	vdc	off
View Name	symbol	off
Instance Name	V0	off

User Property	Master Value	Local Value	Display
Ivignore	TRUE		off

CDF Parameter	Value	Display
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage	1	off
AC magnitude		off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Temperature coefficient 1		off
Temperature coefficient 2		off
Nominal temperature		off



Instance insertion: Resistor

- To complete the schematic for voltage divider, we need to insert resistor and ground:
 - Resistor(Library: “analogLib”, Cell: “res”)
 - Ground (Library: “analogLib”, Cell: “gnd”)

The screenshot shows the 'Edit Object Properties' dialog box for a resistor instance. The 'Apply To' dropdown is set to 'only current' and 'instance'. The 'Show' section has checkboxes for 'system' (unchecked), 'user' (checked), and 'CDF' (checked). Below are buttons for 'Browse' and 'Reset Instance Labels Display'. The main table lists properties: Library Name (analogLib), Cell Name (res), View Name (symbol), and Instance Name (R0). Below this are 'Add', 'Delete', and 'Modify' buttons. The 'CDF Parameter' section lists various parameters: Model name, Resistance (highlighted with a red box and containing '1K Ohms'), Length, Width, Multiplier, Scale factor, Temp rise from ambient, and Temperature coefficient 1. Each parameter has a 'Value' field and a 'Display' dropdown set to 'off'. At the bottom are buttons for 'OK', 'Cancel', 'Apply', 'Defaults', 'Previous', 'Next', and 'Help'.

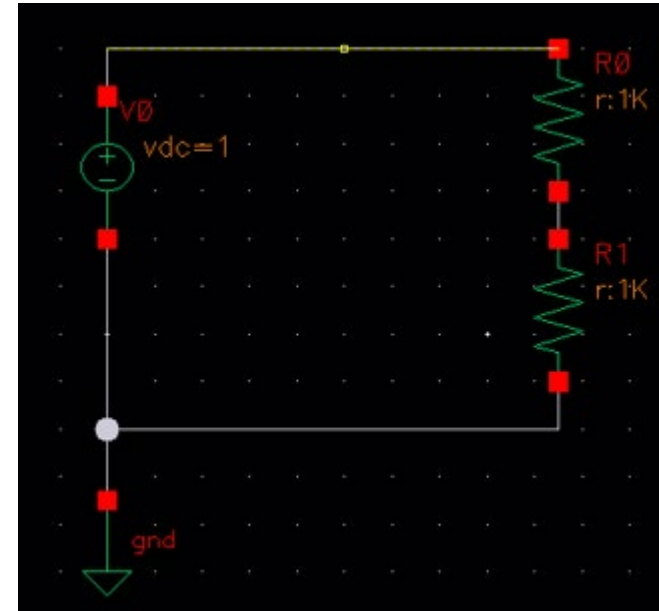
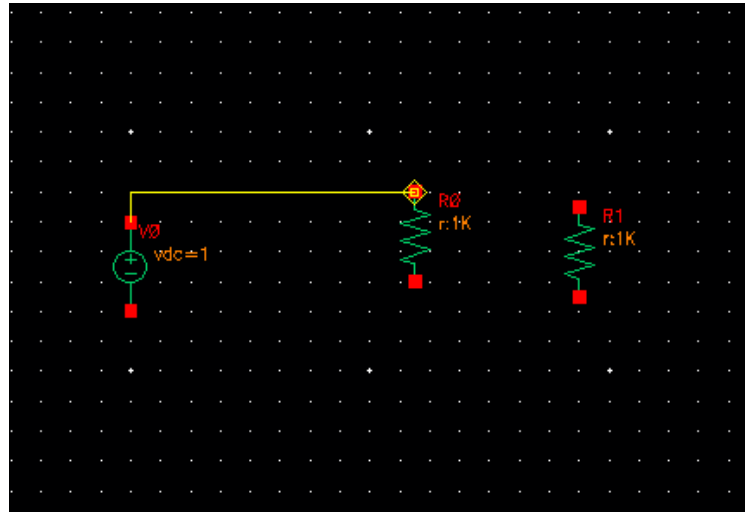
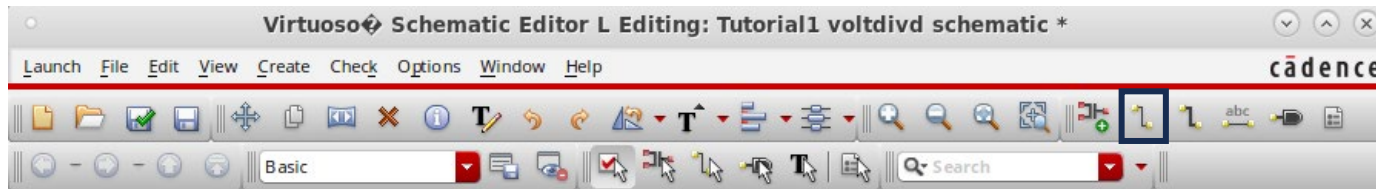
Property	Value	Display
Library Name	analogLib	off
Cell Name	res	off
View Name	symbol	off
Instance Name	R0	off

CDF Parameter	Value	Display
Model name		off
Resistance	1K Ohms	off
Length		off
Width		off
Multiplier		off
Scale factor		off
Temp rise from ambient		off
Temperature coefficient 1		off

- To tune number digit, use the following character:
 - “m”(mili= 10^{-3}), “u”(micro= 10^{-6}), “n”(nano= 10^{-9}), “p”(pico= 10^{-12}), “f”(femto= 10^{-15})
 - “K”(kilo= 10^3), “M”(Mega= 10^6), “G”(Giga= 10^9), “Tera”(Tera= 10^{12})

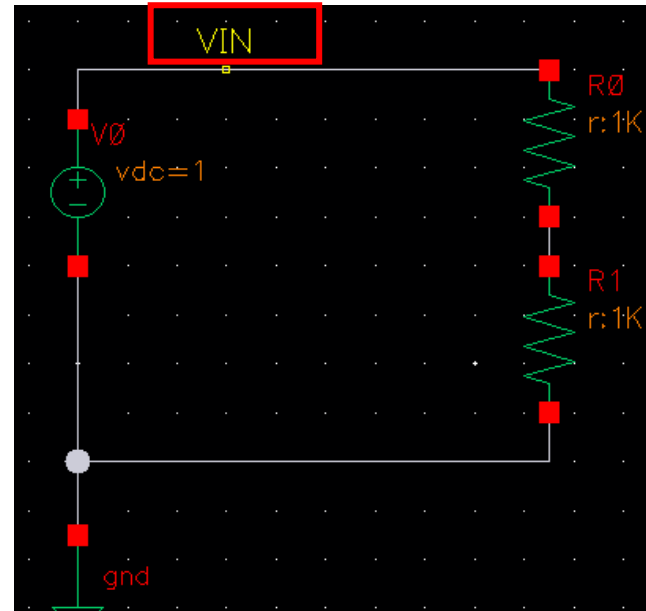
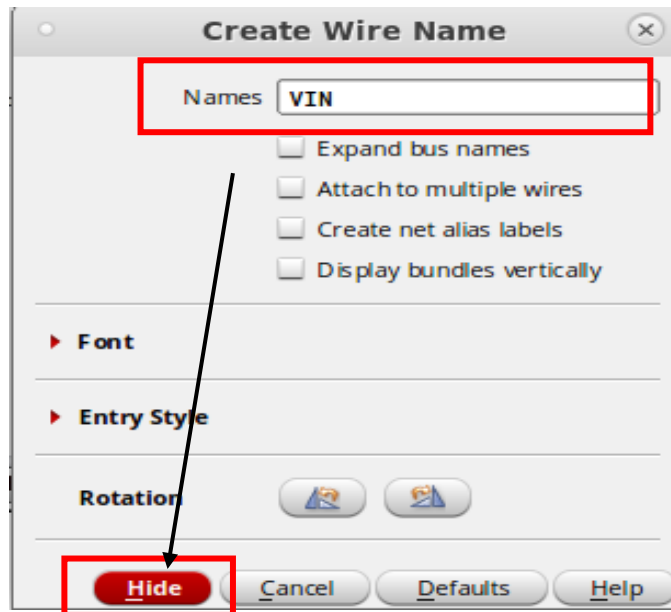
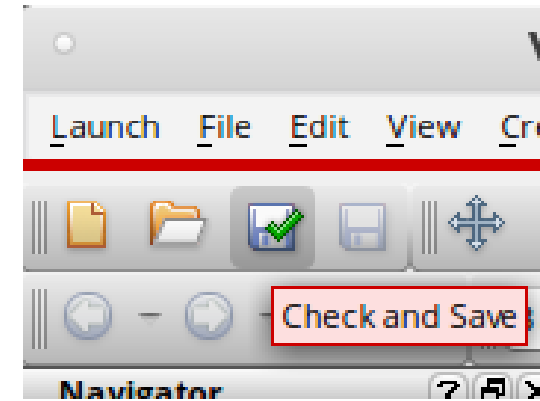
Wiring

- In order to connect two node electrically, we need to wire them together.
- To wire, click "w", or icon. 
- Connect the two nodes by simply clicking both nodes.



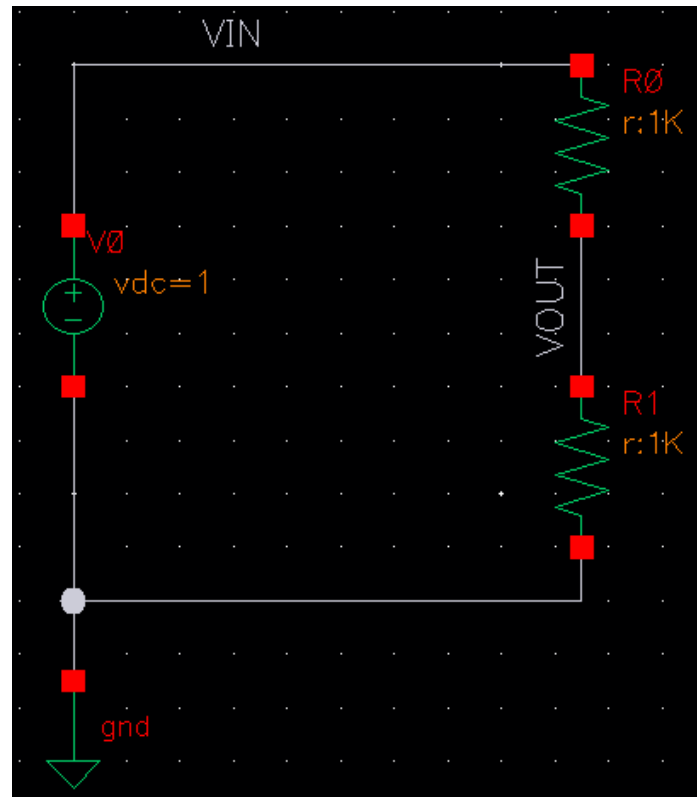
Net label

- Click "l" (L for label), a window **Create Wire Name** popup.
- Type "VIN" for Name, then click "Hide" to display the label.
- A text "VIN" will stick to your cursor, place it on the net
- After label, click **File>Check and Save**



Completion of schematic

- Complete your circuit label as shown, remember to save.
- Now schematic for voltage divider is completed and ready for simulation.



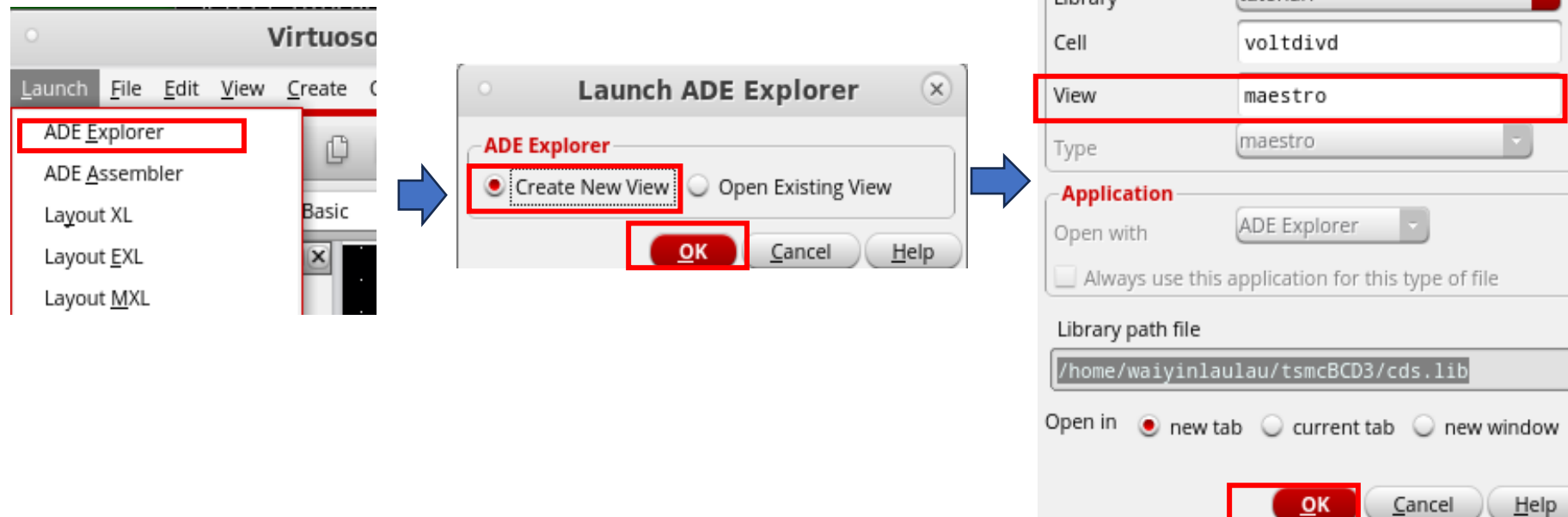
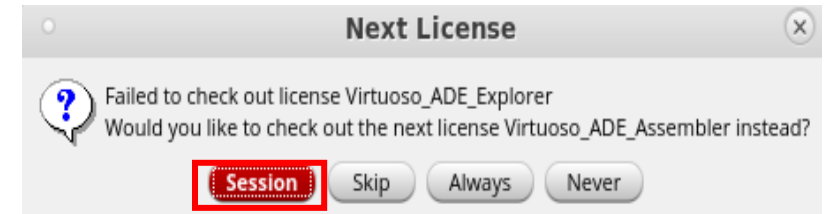
Exercise2: DC Simulation

Action Item:

- Open ADE Explorer,
- Set DC analysis
- Run simulation

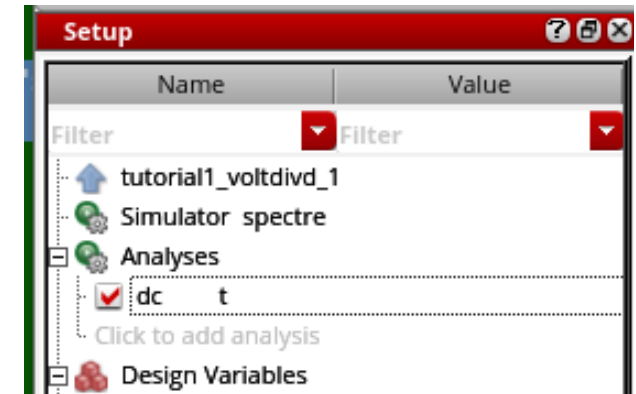
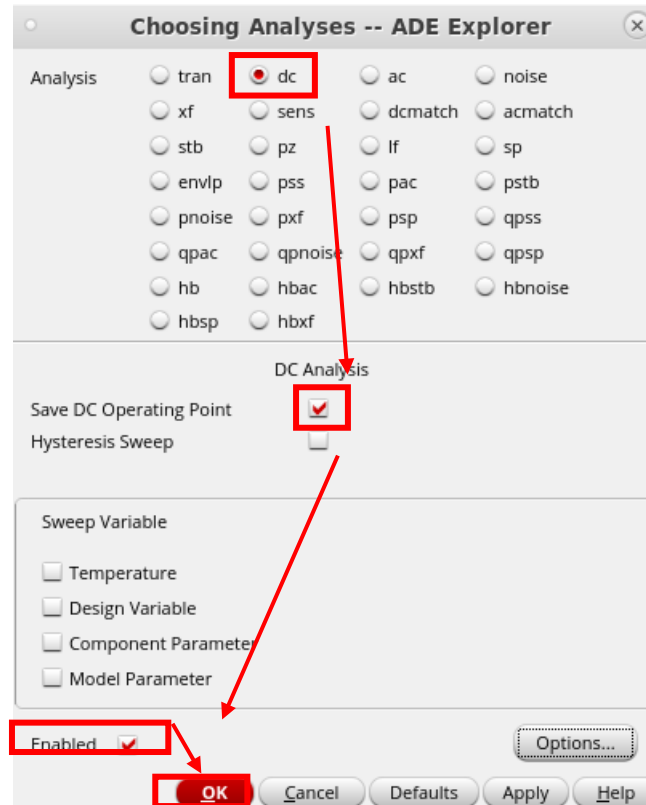
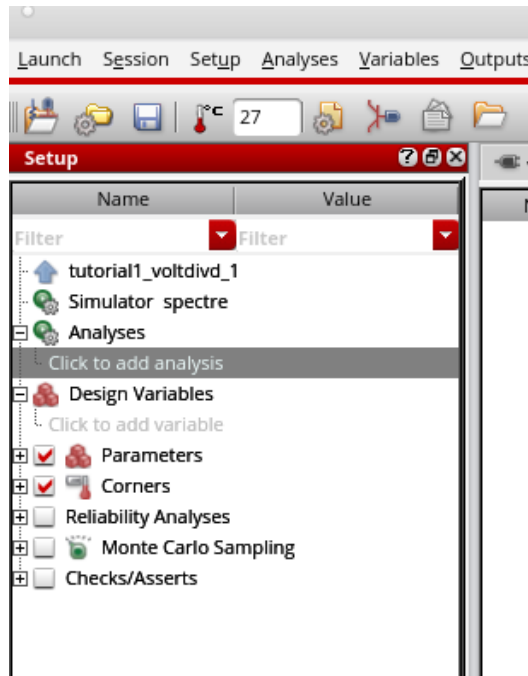
Start Simulation: ADE Explorer

- Click **launch>ADE Explorer** from schematic editor window
- If a Next License window appears as shown, click **Session**
- **ADE Explorer** window pops up, we will adjust any simulation-related setting here



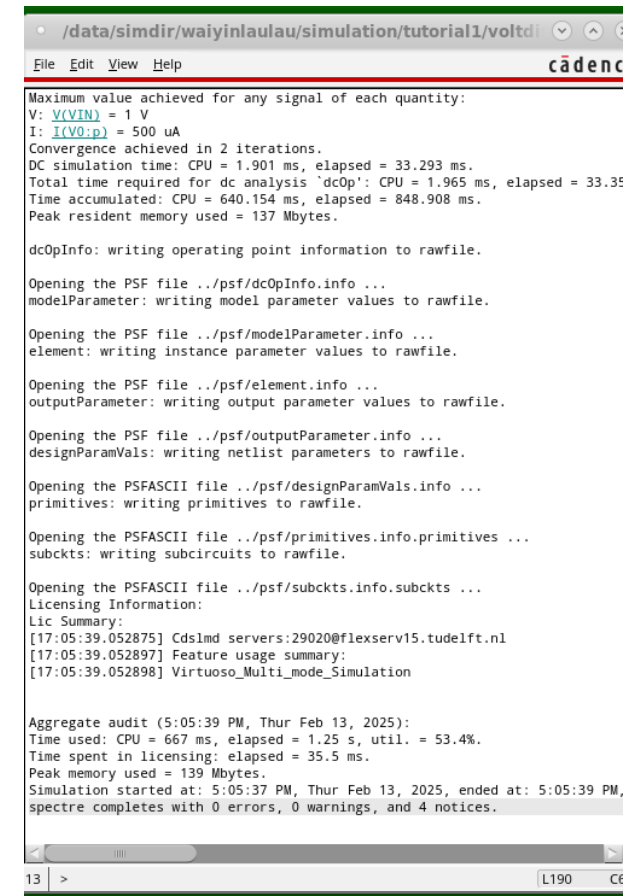
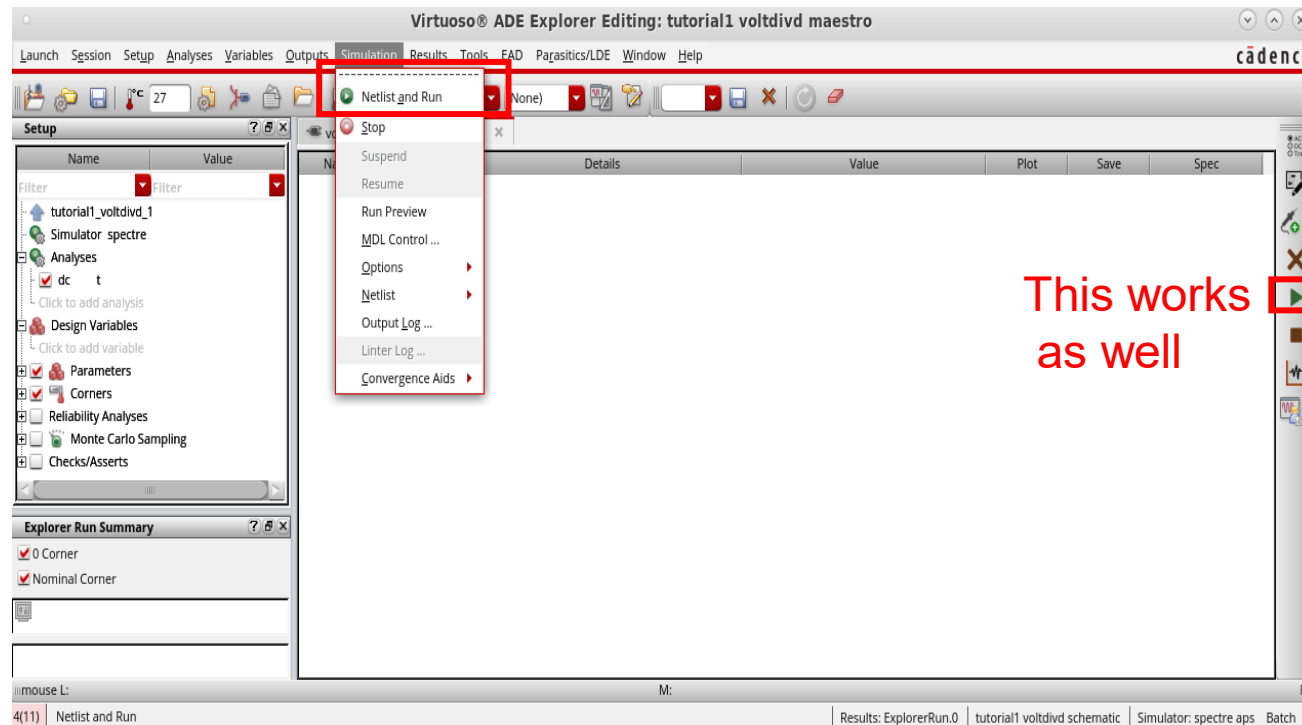
ADE Explorer: DC analysis setting

- Select **DC** under Analysis
- Click **Save DC operating point**



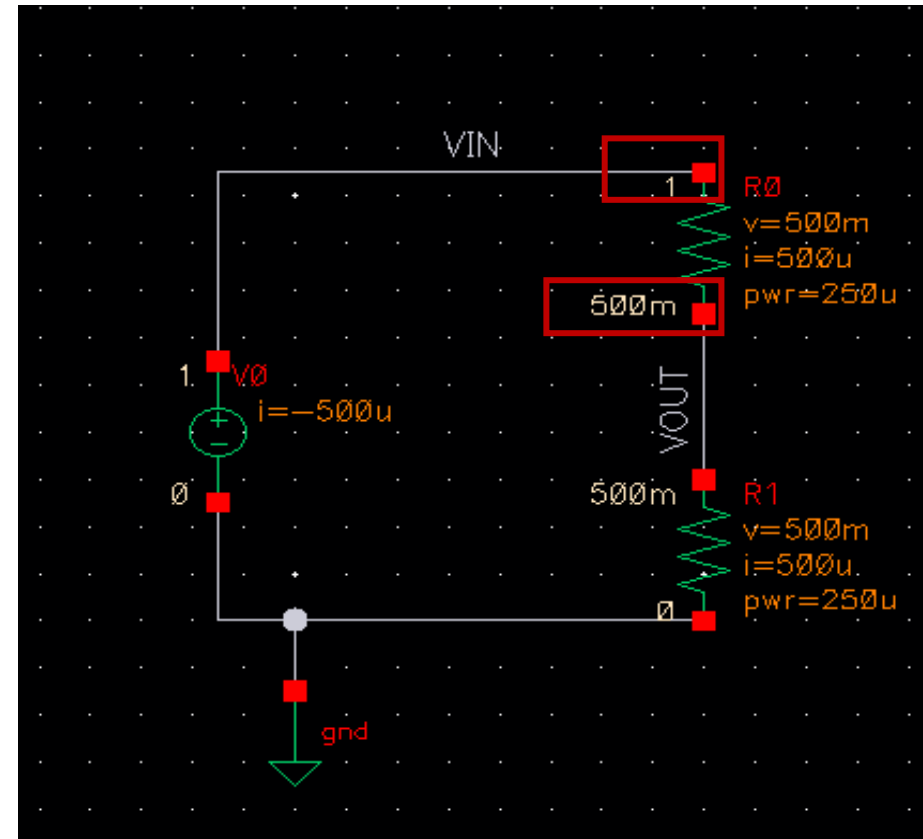
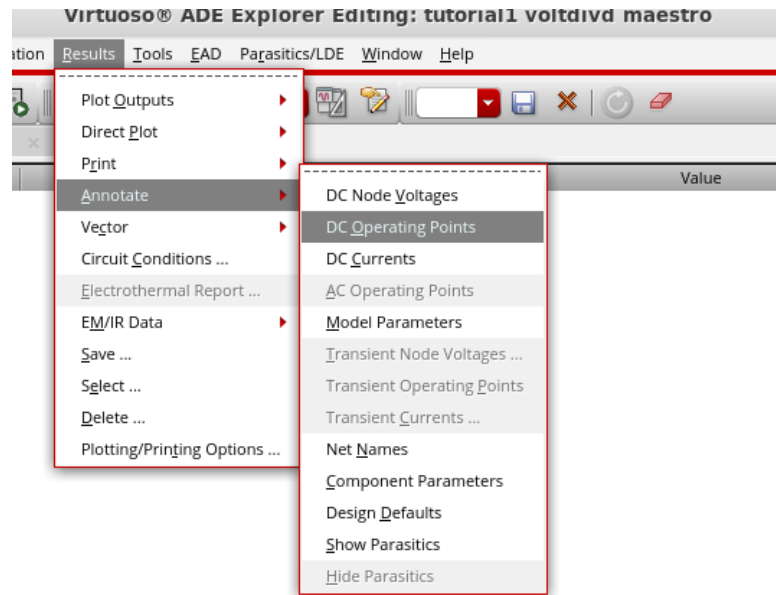
ADE Explorer : Run simulation

- Click icon **Netlist and Run** to run simulation
- After simulation successfully, there will be a window pop-up, stating simulation information
- In case of an error, check the **CIS window (CDS.log)**



ADE Explorer : Parameter Annotate

- In order to display value on the schematic directly, go to **Results>Annotate>DC Node Voltage**
- After Exercise2, the simulation result is like this



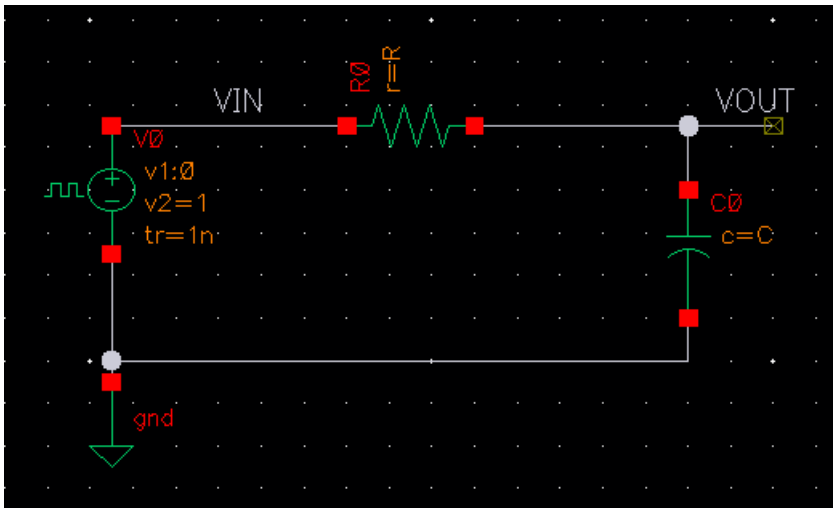
Exercise3:Transient simulation

Action Item:

- Complete a RC-filter schematic
- Open the ADE Explorer for simulation
- Set the transient analysis and run simulation

Tran simulation: RC-filter

- Complete a schematic as shown.
 - Pulse voltage source (Library: “analogLib”, Cell: “vpulse”)
 - Capacitor (Library: “analogLib”, Cell: “cap”)
 - Change the value as shown. We use “R”, and “C” for resistance and capacitance, that making both values become “design variable”.



Library Name	analogLib
Cell Name	res
View Name	symbol
Instance Name	R0

Add Delete Modify

CDF Parameter	Value
Model name	
Resistance	R Ohms

Library Name	analogLib
Cell Name	cap
View Name	symbol
Instance Name	C0

Add Delete Modify

CDF Parameter	Value
Model name	
Capacitance	C F
Width	

Library Name	analogLib
Cell Name	vpulse
View Name	symbol
Instance Name	V0

Add Delete Modify

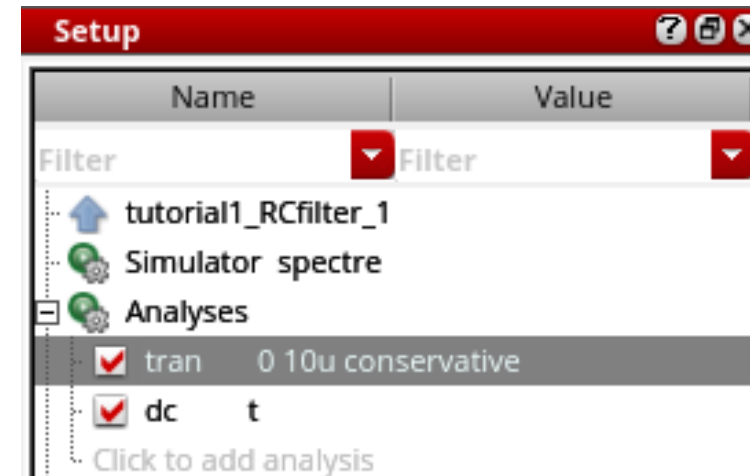
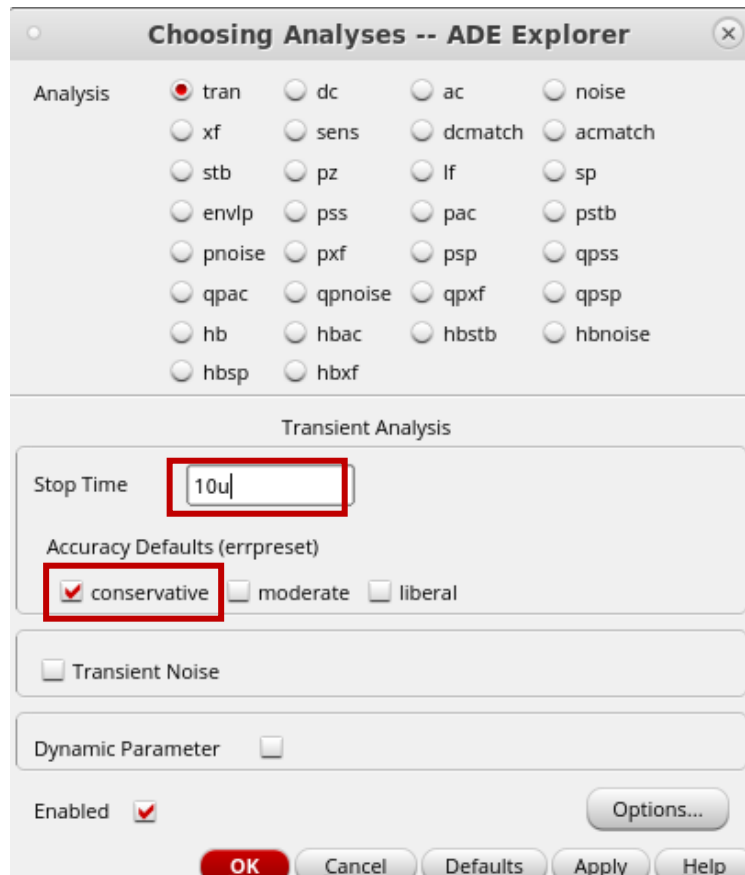
User Property Master Value Local Value

IvIgnore TRUE

CDF Parameter	Value
Frequency name for 1/period	
Noise file name	
Number of noise/freq pairs	0
DC voltage	
AC magnitude	
AC phase	
XF magnitude	
PAC magnitude	
PAC phase	
Voltage 1	0 V
Voltage 2	1 V
Period	1 u s
Delay time	
Rise time	1 n s
Fall time	1 n s

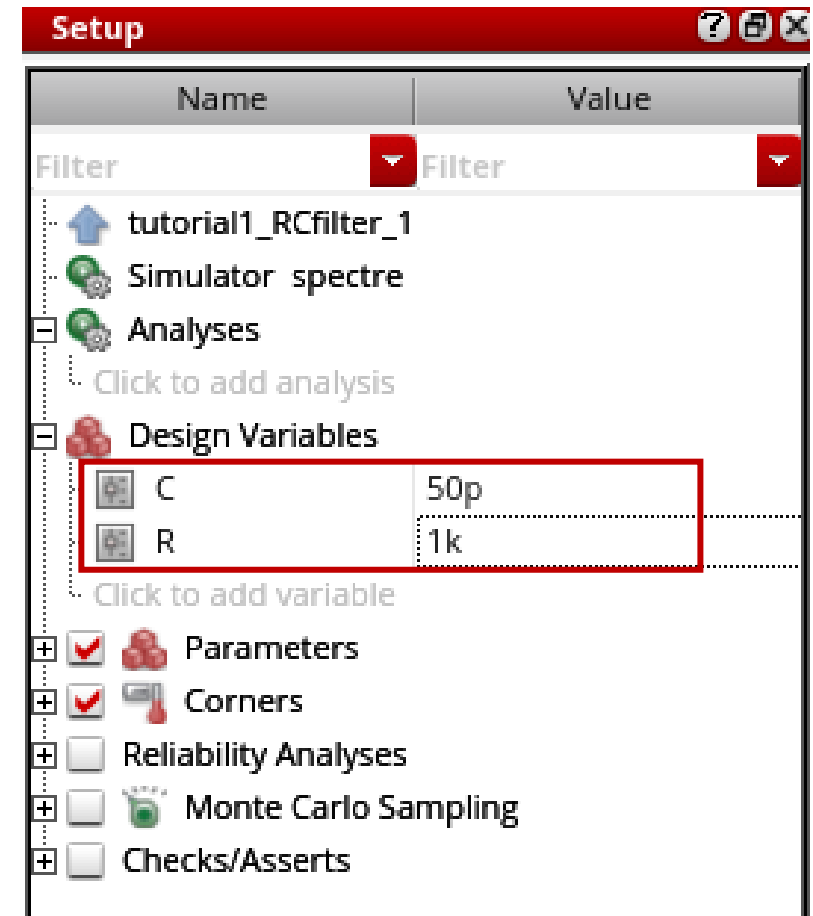
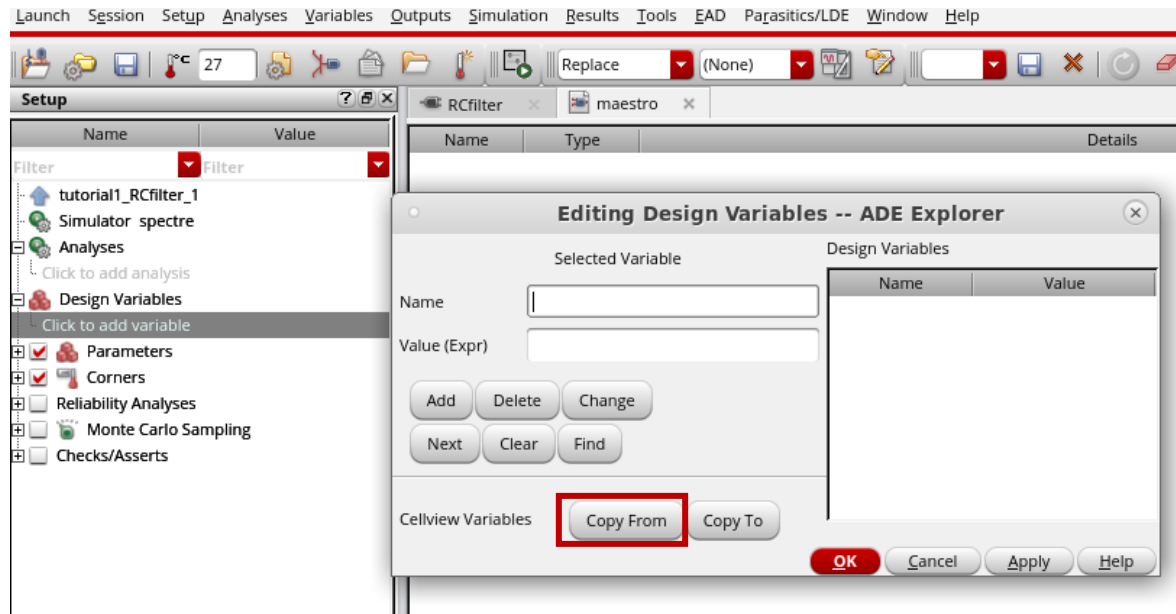
Choose Transient Analysis

- First, choose **DC** analysis to save a DC operating point.
- Then choose **tran** analysis, set stop time to be 10us
- Choose **conservative** as Accuracy



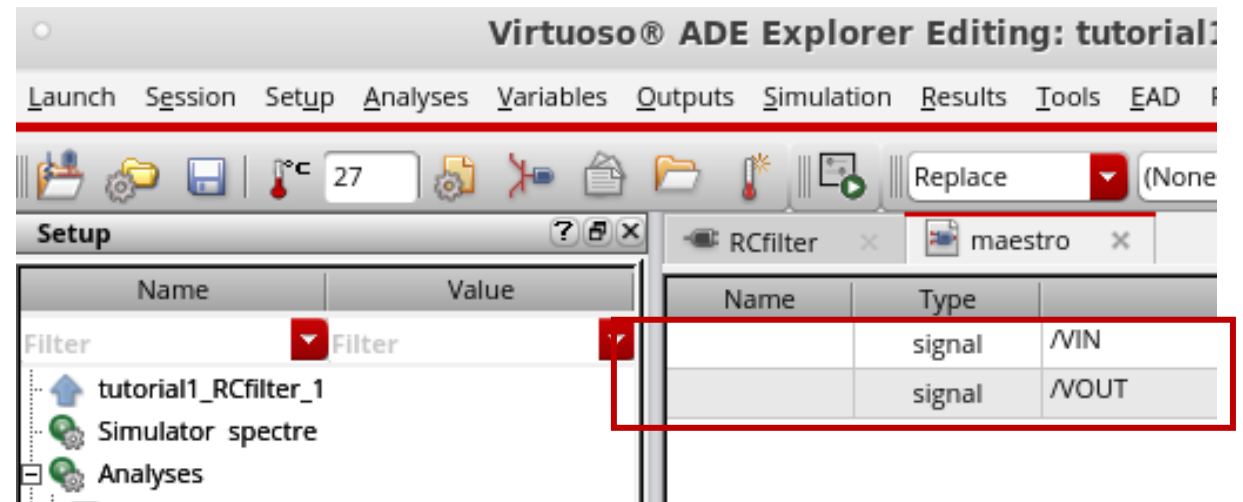
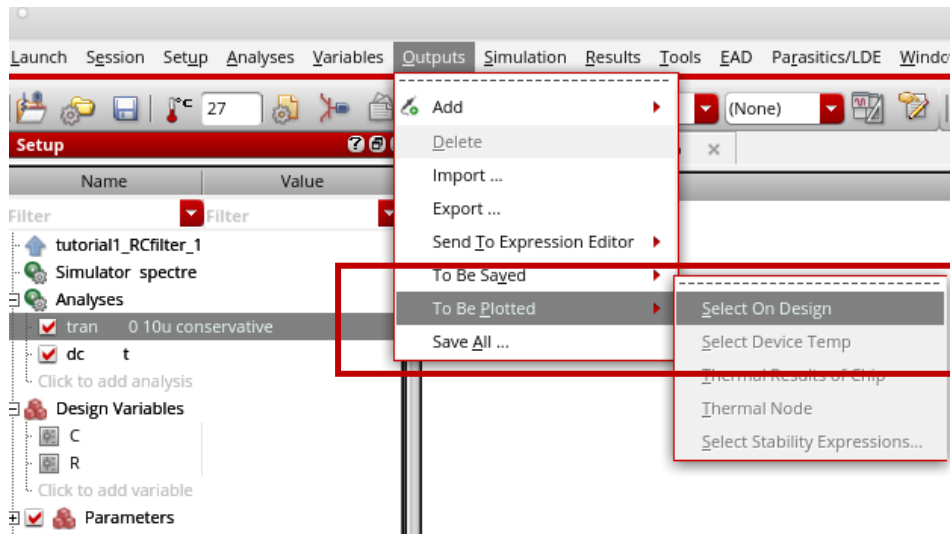
Set Design Variables

- Here we set Design Variable R and C
- Modify their value as R=1K and C=50p

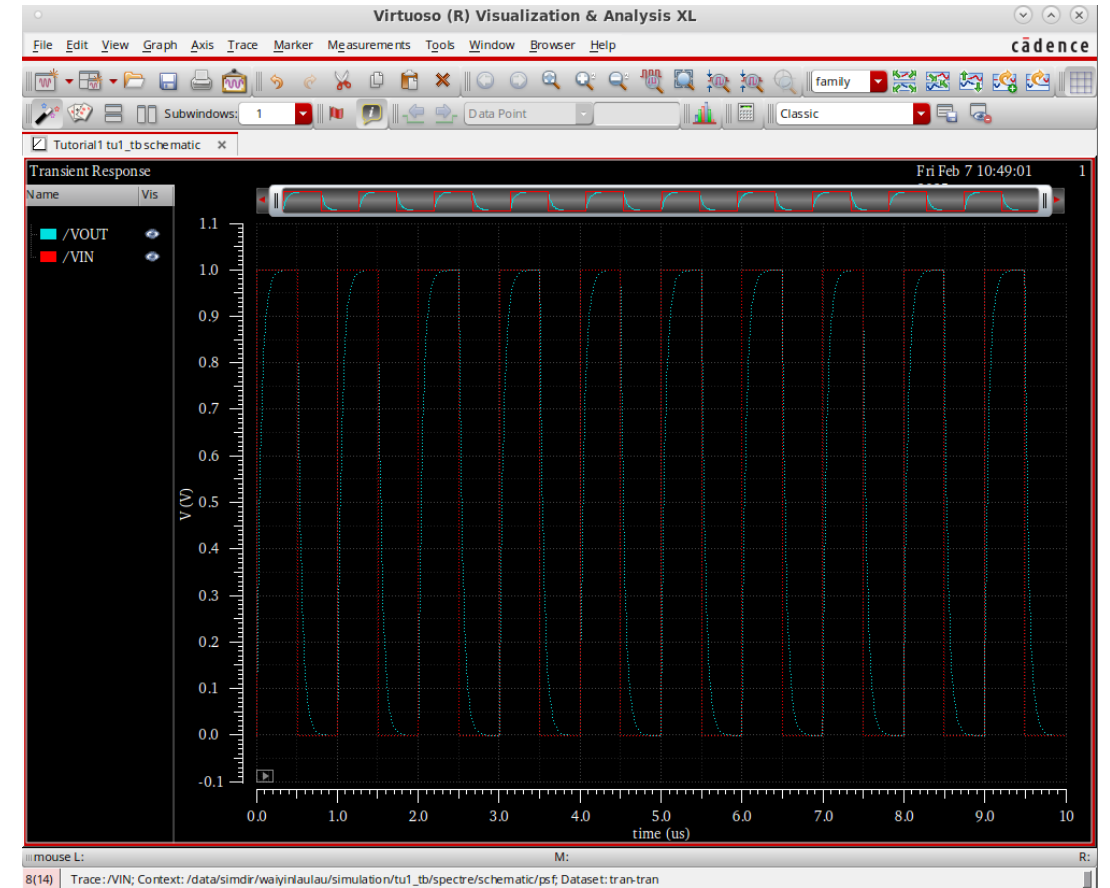
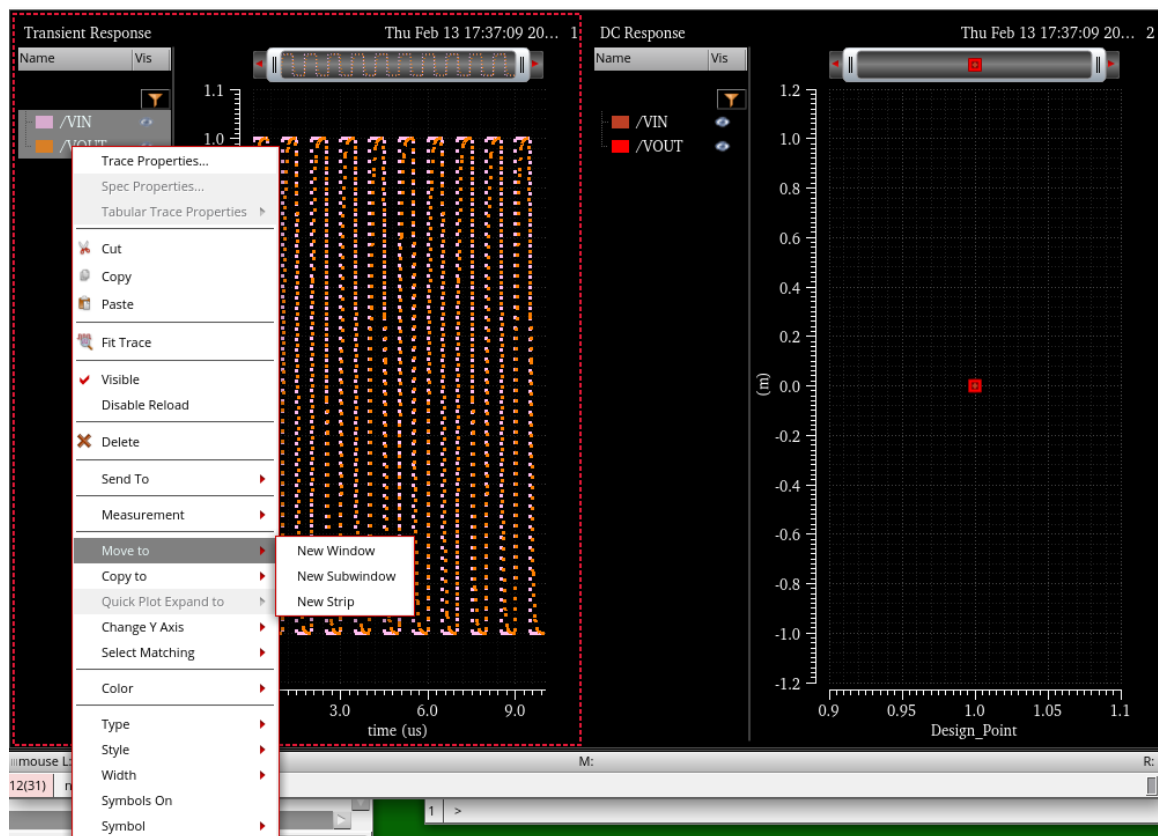


Sim setting: Output selection

- In ADE explorer window, go to **Outputs>To Be Plotted>Select On Design**.
- It will guide you back to your schematic tab, click VIN and VOUT
- Back to the ADE Explorer mastero tab, both VIN and VOUT are shown in Outputs
- The transient simulation is now ready. Click **Netlist and Run** to run the simulation.

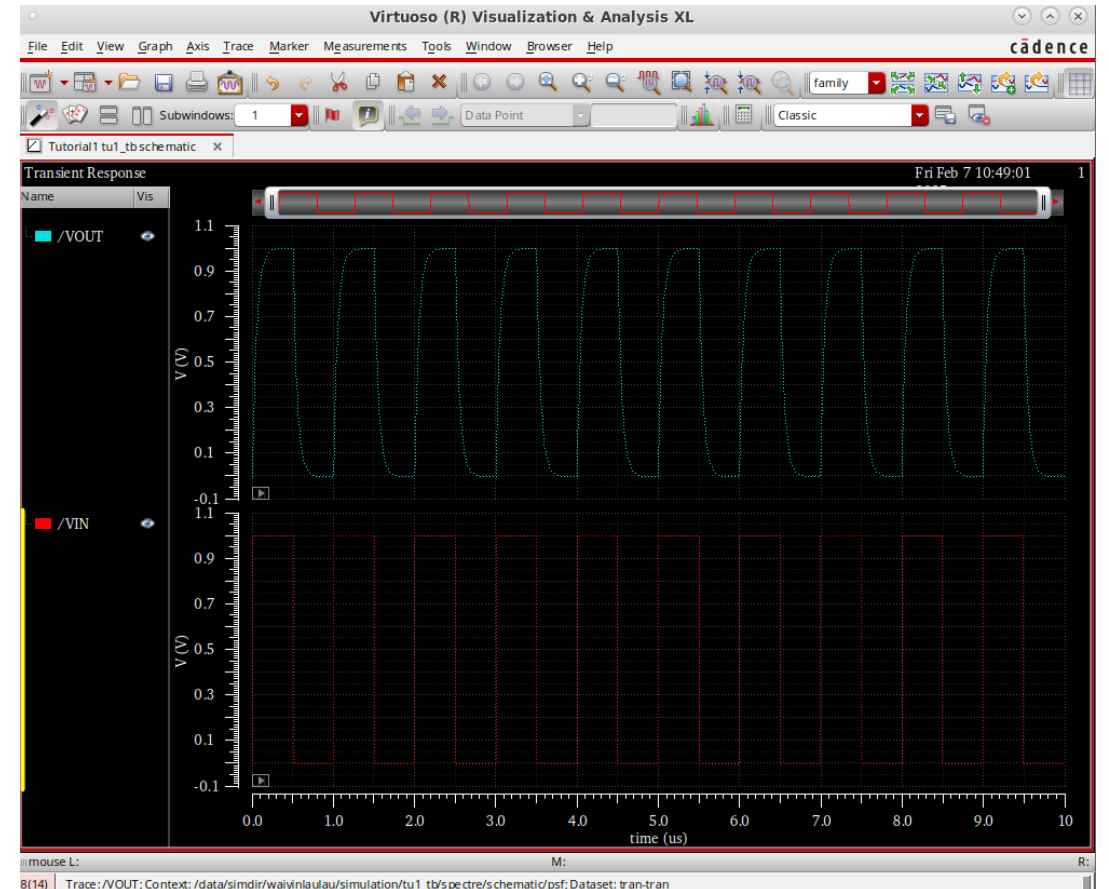
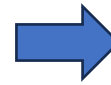
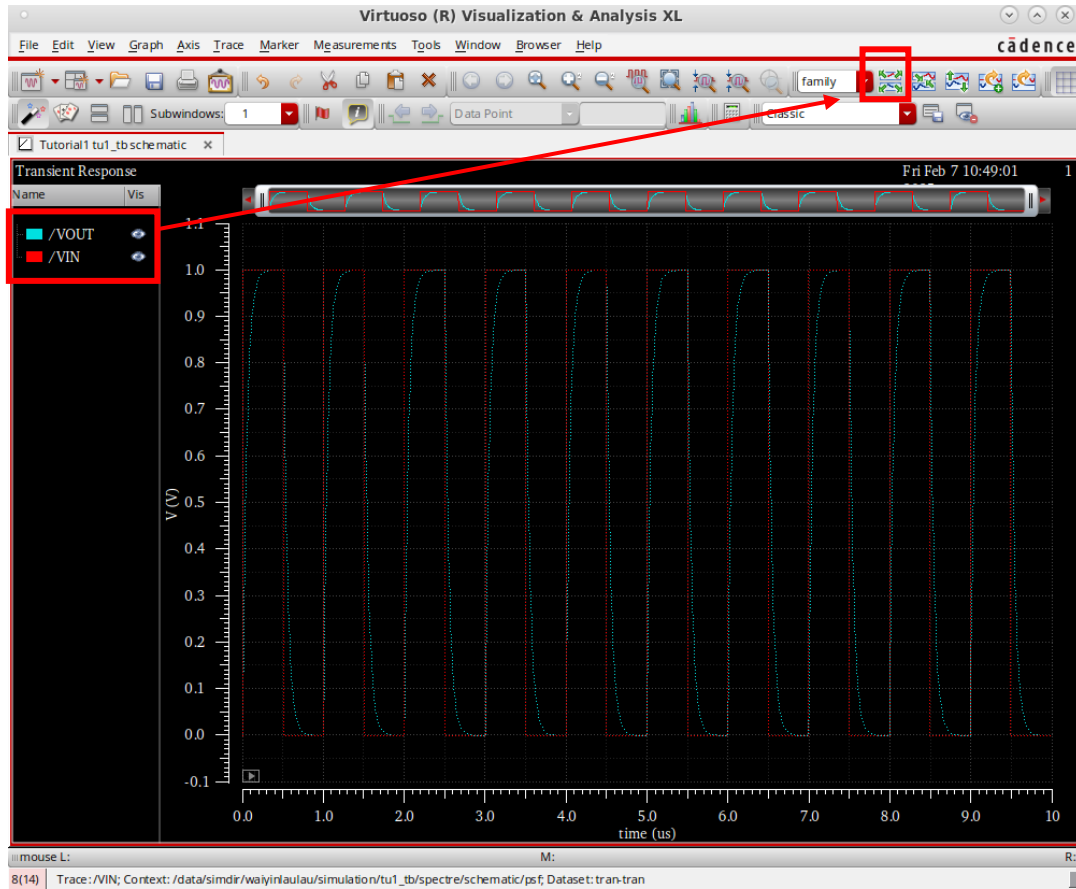


Tran simulation: Output plot(1)



Tran simulation: Output plot(2)

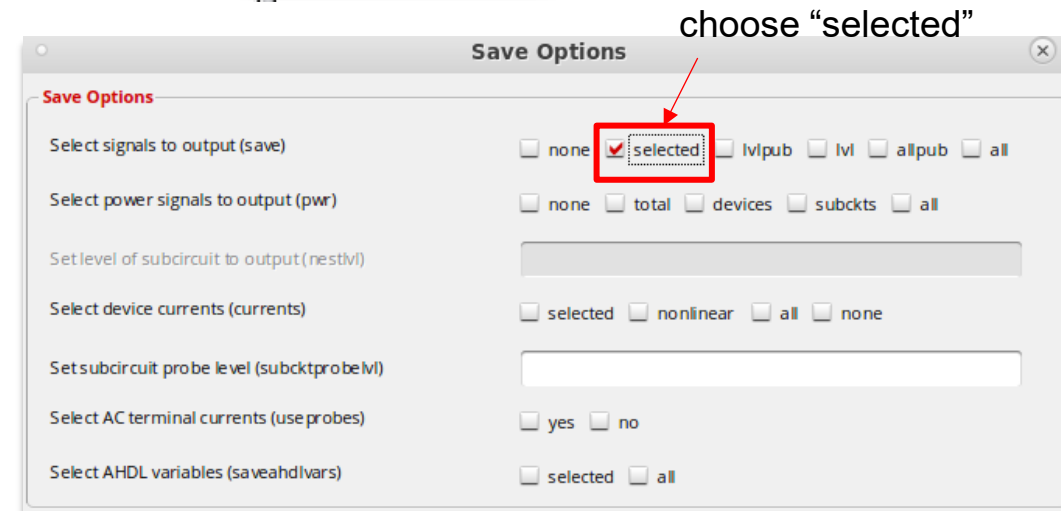
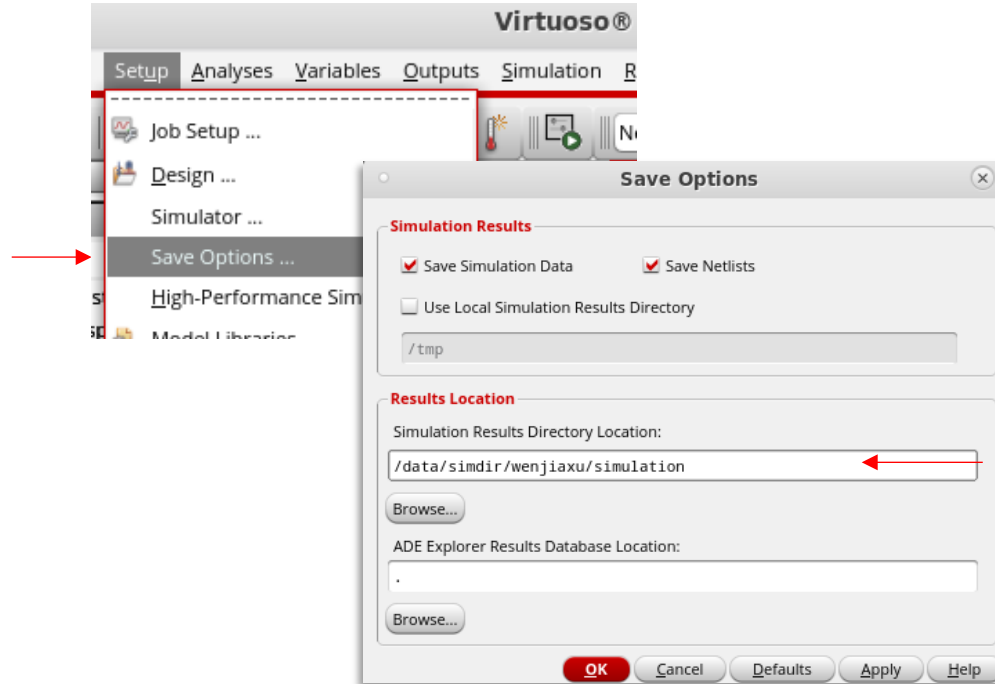
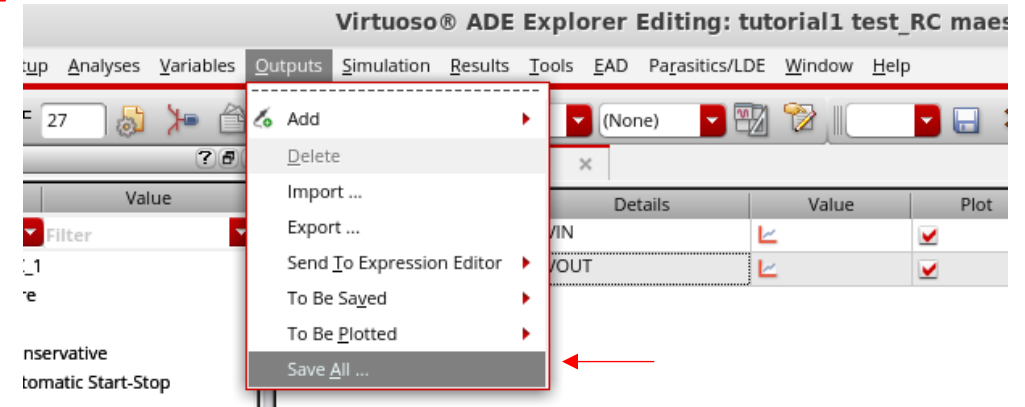
- Select the waveform then click Split All Strips to separate them



Save Option

To limit the simulation size and avoid crash:

1. Save Options: choose “selected” to store only selected nodes
2. Check your simulation result directory in Setup>Save Options...
directory should be “/data/simdir/your username/simulation”
3. You can delete previous simulation file by re-run the simulation and stop immediately, this will overwrite the previous large one.

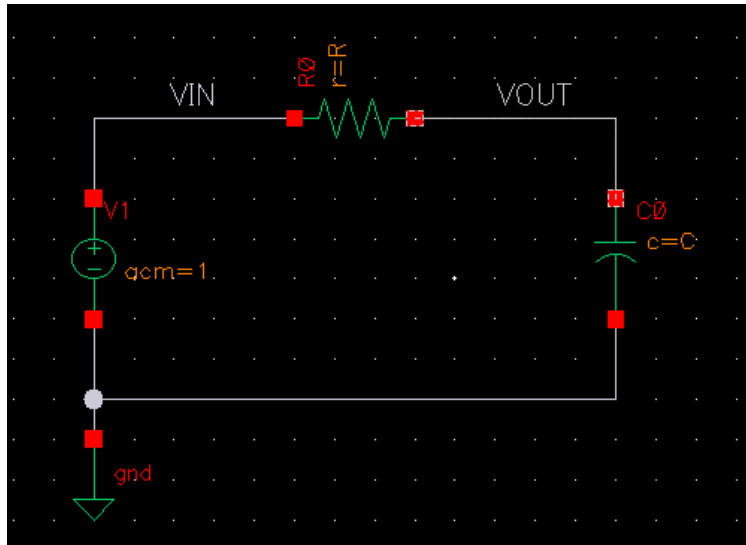


“/data/simdir/your username/simulation”

AC simulation

Schematic:

- Added instance “vdc”
- Hit hotkey “q” to edit properties
- Change “AC magnitude” to 1, “OK”
- Click “OK”, the voltage source will show “acm=1” now.



The 'Edit Object Properties' dialog box is shown. It has a title bar with a close button. The 'Apply To' section has 'only current' and 'instance' buttons, with 'instance' selected. The 'Show' section has checkboxes for 'system', 'user', and 'CDF', with 'user' and 'CDF' checked. Below these are 'Browse' and 'Reset Instance Labels Display' buttons. The main table has columns for 'Property', 'Value', and 'Display'. The 'Property' column lists 'Library Name', 'Cell Name', 'View Name', and 'Instance Name'. The 'Value' column shows 'analogLib', 'vdc', 'symbol', and 'V1'. The 'Display' column shows 'off' for all. Below this is a section for 'User Property' with 'Add', 'Delete', and 'Modify' buttons. The 'User Property' table has columns for 'Property', 'Master Value', 'Local Value', and 'Display'. The 'Property' column lists 'Ivsignore'. The 'Master Value' column shows 'TRUE'. The 'Local Value' column is empty. The 'Display' column shows 'off'. Below this is a section for 'CDF Parameter' with columns for 'Property', 'Value', and 'Display'. The 'Property' column lists 'Noise file name', 'Number of noise/freq pairs', 'DC voltage', 'AC magnitude', 'AC phase', 'XF magnitude', 'PAC magnitude', 'PAC phase', 'Temperature coefficient 1', 'Temperature coefficient 2', and 'Nominal temperature'. The 'Value' column shows empty fields for most, '0' for 'Number of noise/freq pairs', and '1 V' for 'AC magnitude'. The 'Display' column shows 'off' for all. A red arrow points to the 'AC magnitude' field. At the bottom are 'OK', 'Cancel', 'Apply', 'Defaults', 'Previous', 'Next', and 'Help' buttons.

Property	Value	Display
Library Name	analogLib	off
Cell Name	vdc	off
View Name	symbol	off
Instance Name	V1	off

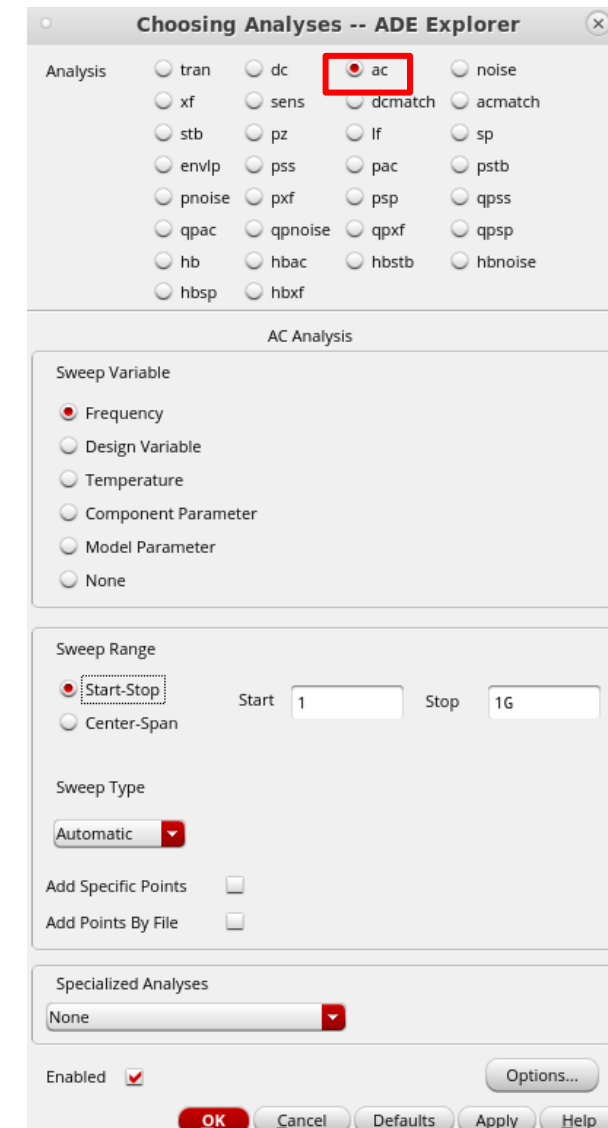
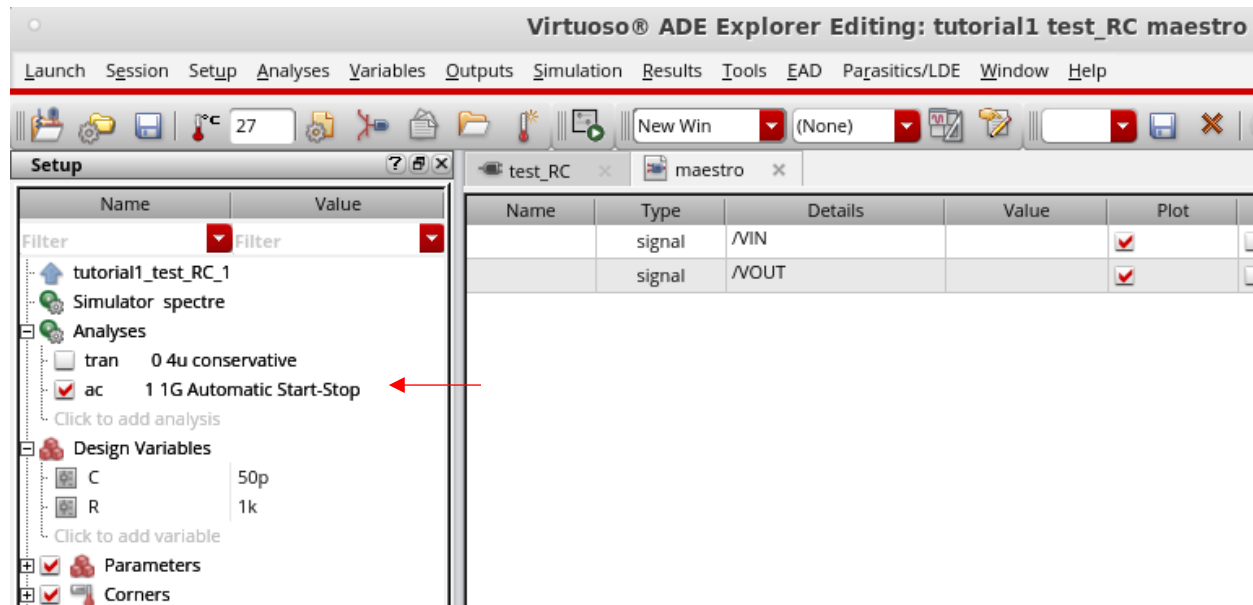
User Property	Master Value	Local Value	Display
Ivsignore	TRUE		off

CDF Parameter	Value	Display
Noise file name		off
Number of noise/freq pairs	0	off
DC voltage		off
AC magnitude	1 V	off
AC phase		off
XF magnitude		off
PAC magnitude		off
PAC phase		off
Temperature coefficient 1		off
Temperature coefficient 2		off
Nominal temperature		off

AC simulation

ADE L:

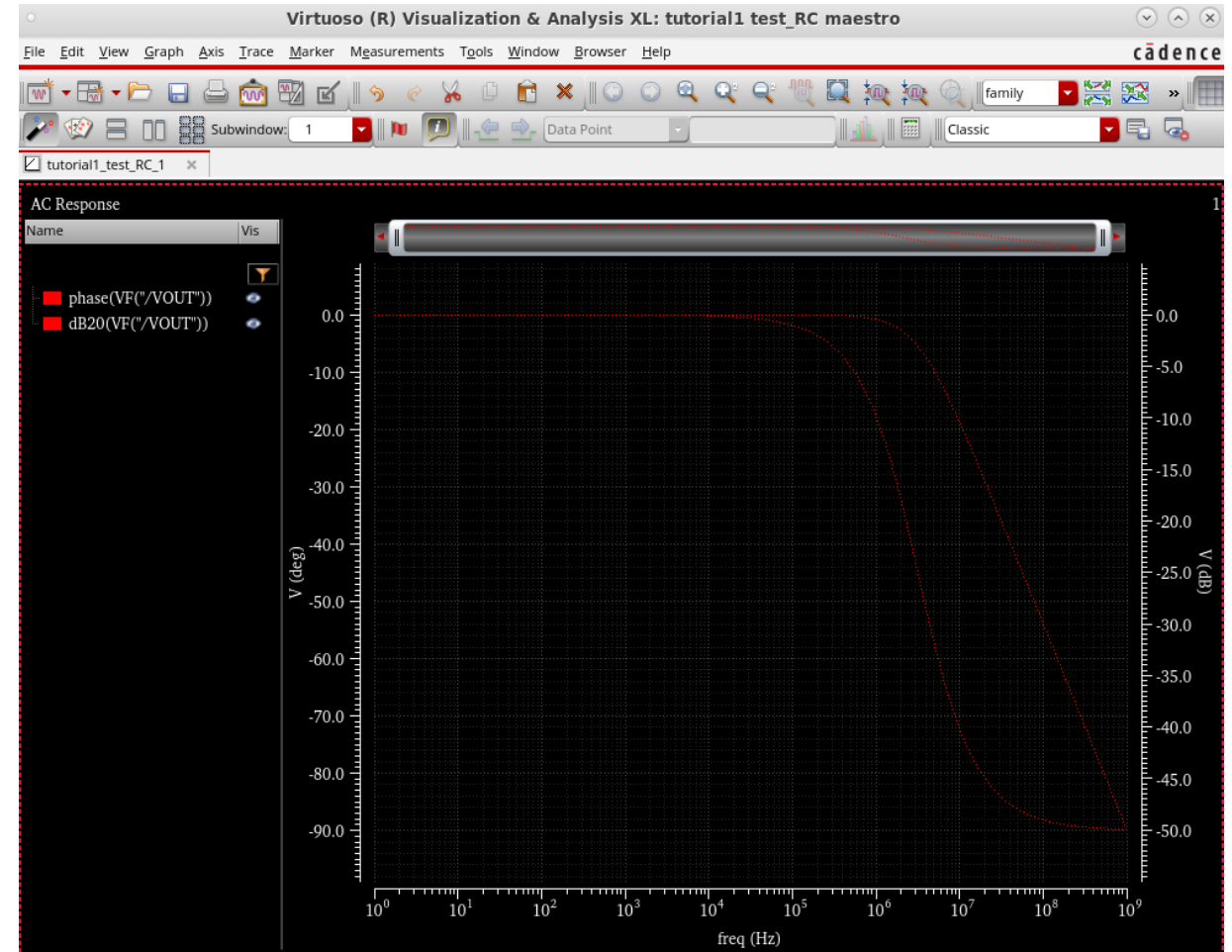
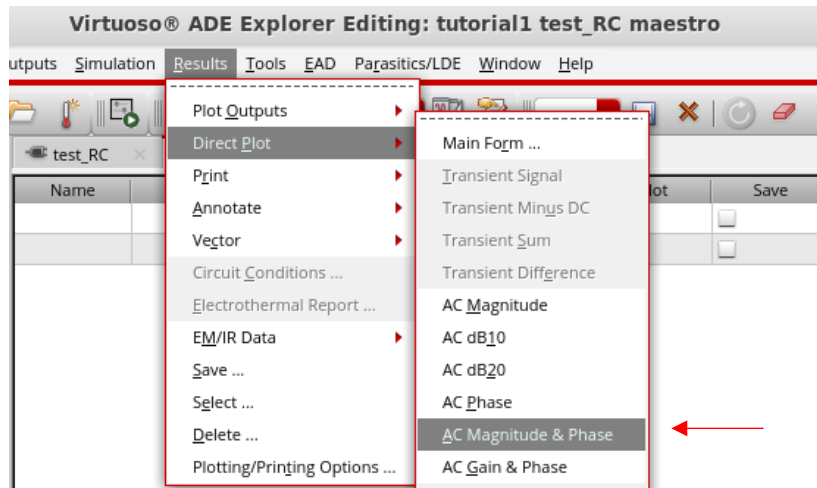
- Analyses-> click to add analysis
- Select “ac” in Analysis.
- Set the frequency range from 1 to 1G and click “OK”.
- Uncheck other simulations to run the ac simulation only



AC simulation

ADE L:

- Run AC simulation
- To view the frequency response,
 - Results -> Direct Plot -> AC Magnitude and Phase.
 - Select “VOUT” net on schematic with left-click and press “Esc”.
 - Then the phase and magnitude response should appear in ViVA

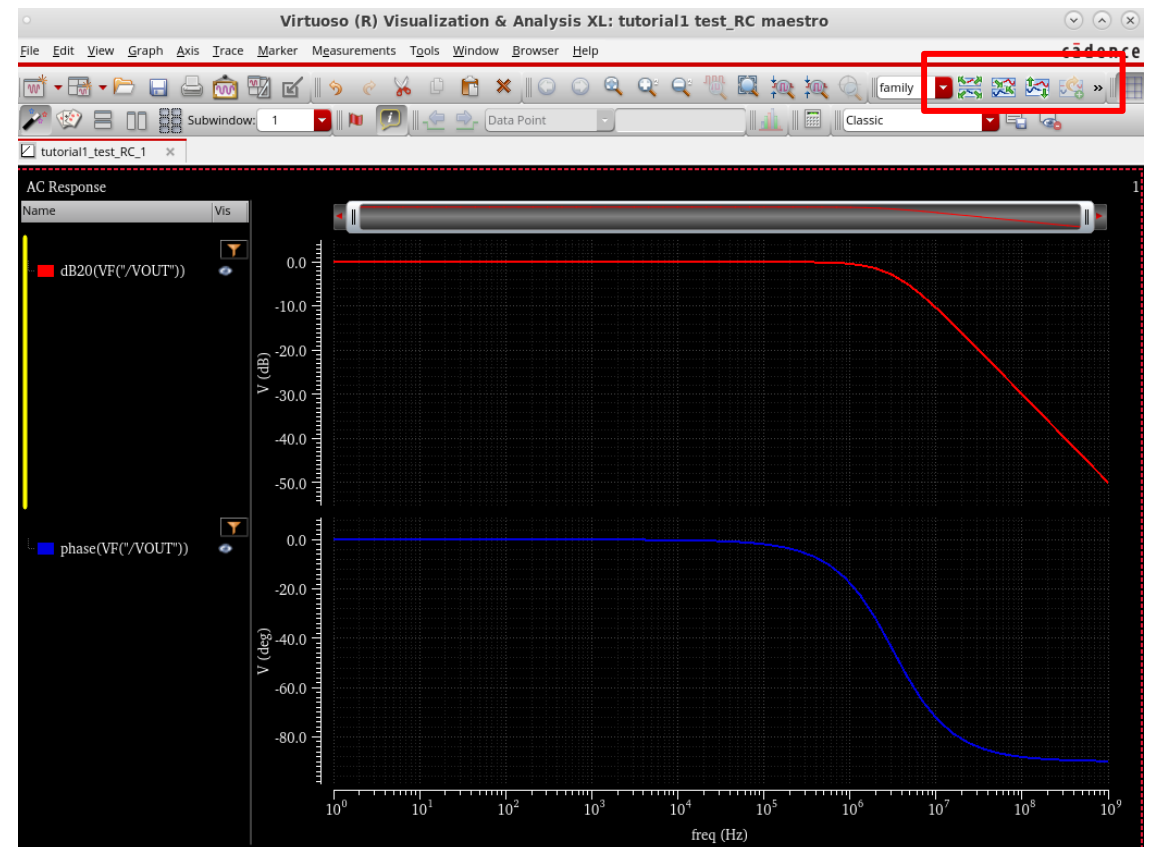
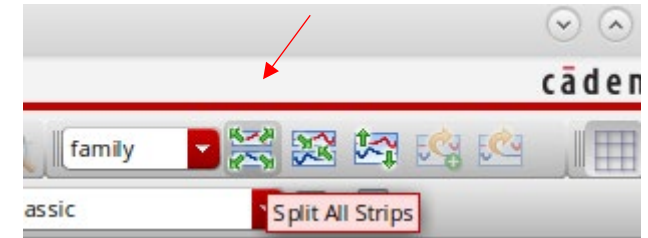
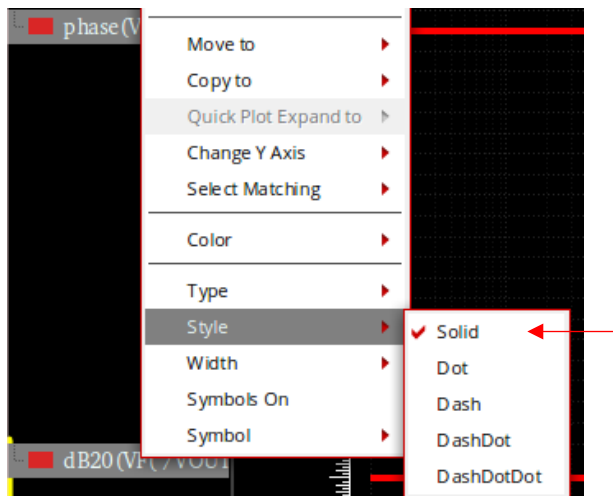


This graph is not clear enough.

Plotting a waveform

ViVA:

- Click “Split All Strips” at the top. You can adjust their sequence by drag and drop.
- Change line Width/Style/Color
 - Select wires(ctrl), right click -> Style ->Solid; Width> Medium
- You can create markers on waveform by pressing:
 - “A” “B” for Differential marker, “M” for Point marker
 - “V” “H” for Vertical and Horizontal marker

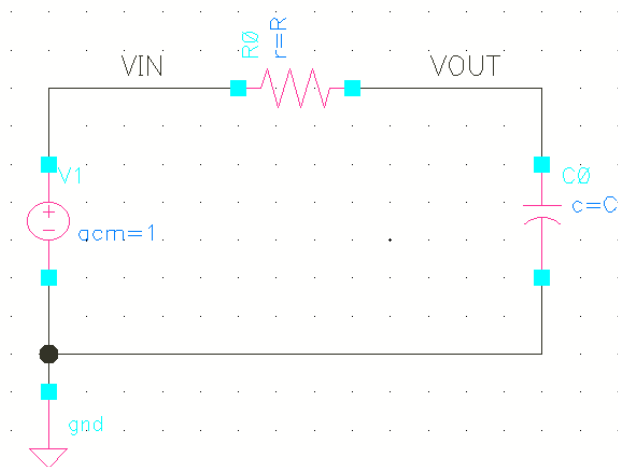


Saving Schematic as image

- Please use a white background in your presentation and homework
- Size the window to the proper size and aspect ratio for better image quality
- The critical device parameters (R, C, W, L, etc.) should be visible

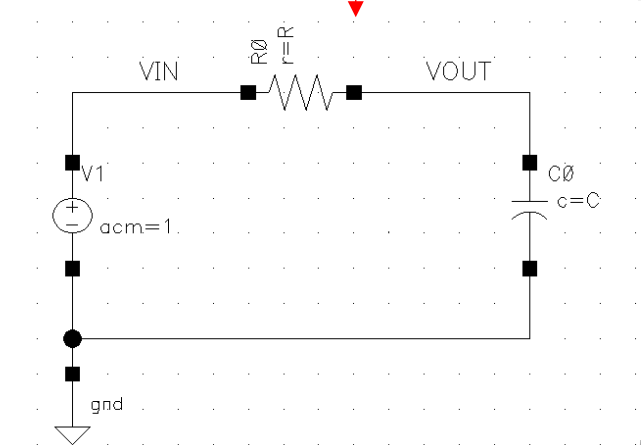
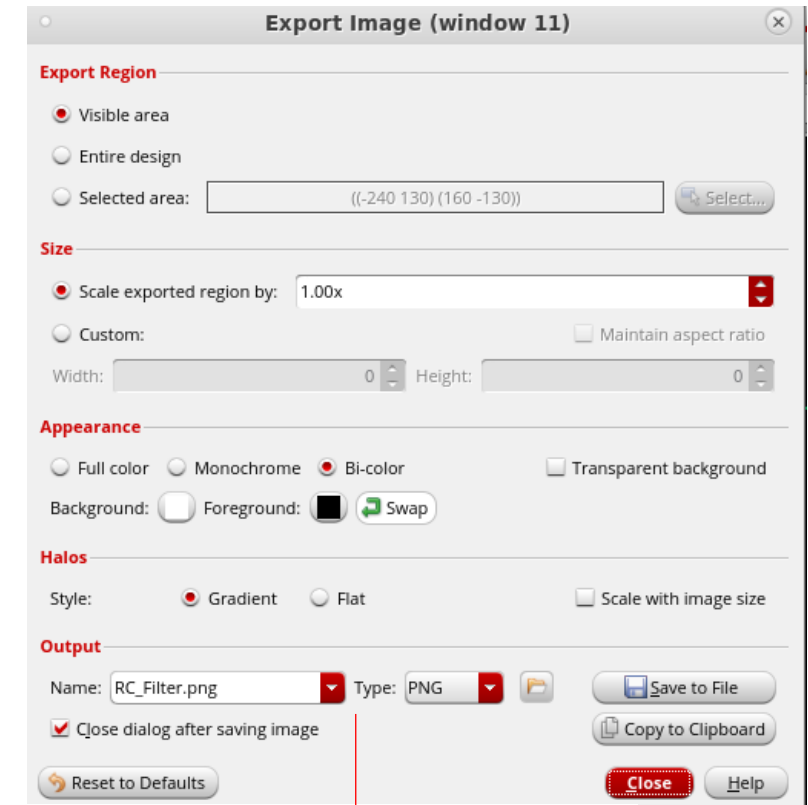
Tips for schematic:

- “F” for whole , “Right click + drag” to zoom in essential part.
- Take a screenshot, simply use “Shift+Win+S” from Windows, or “Shift-Command-4” in Mac
Then use Paint App to invert the image color;
- Or export schematic image as white background, then take a screenshot.



white background
like this

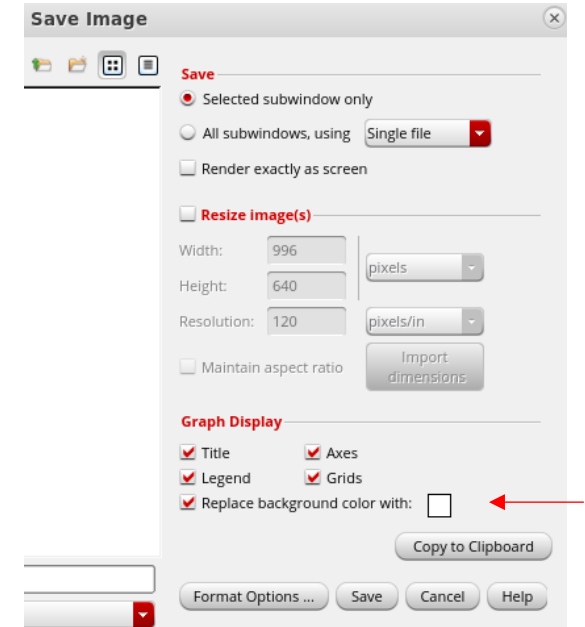
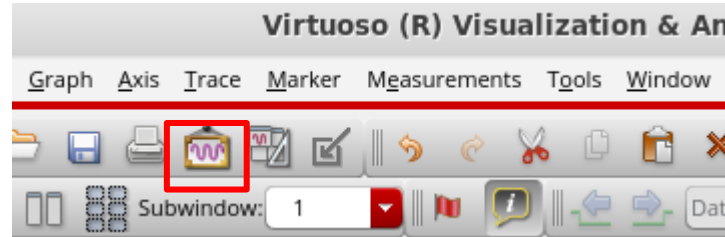
Or this



Saving Waveform as image

Tips for waveform:

- Save waveform with white background:



- You can also set viva background to white by add this command into ".cdsinit" file:
 - `envSetVal("viva.graphFrame" "background" 'string "white")`

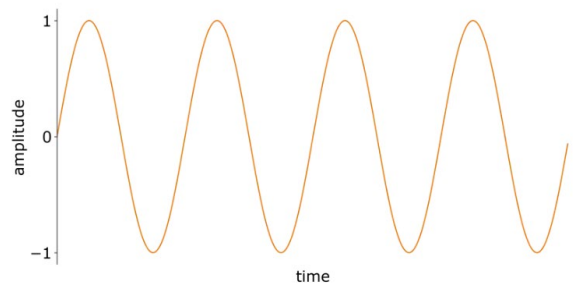


To find ".cdsinit" file, click on "show hidden files" first.

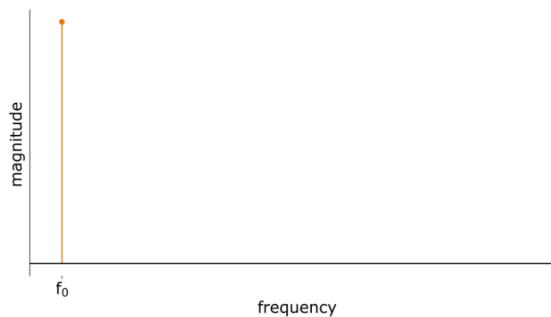
- Then take a screenshot.

FFT(Fast Fourier Transform)

- Time domain

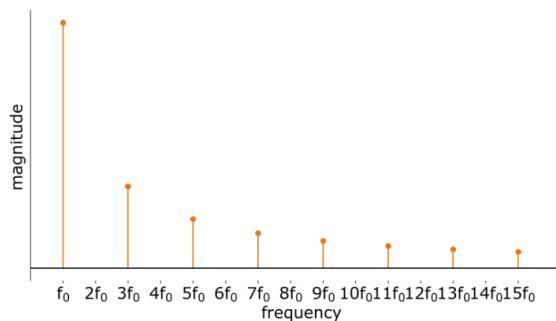
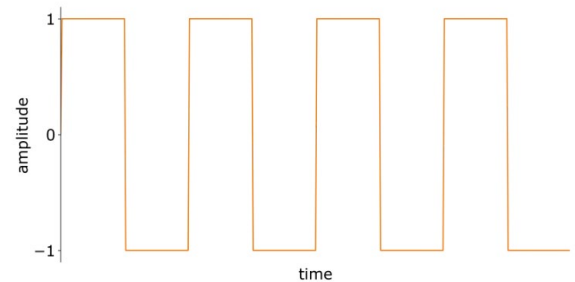


- Frequency domain



What can we see from fft spectrum?

- Distortion
- Frequency component



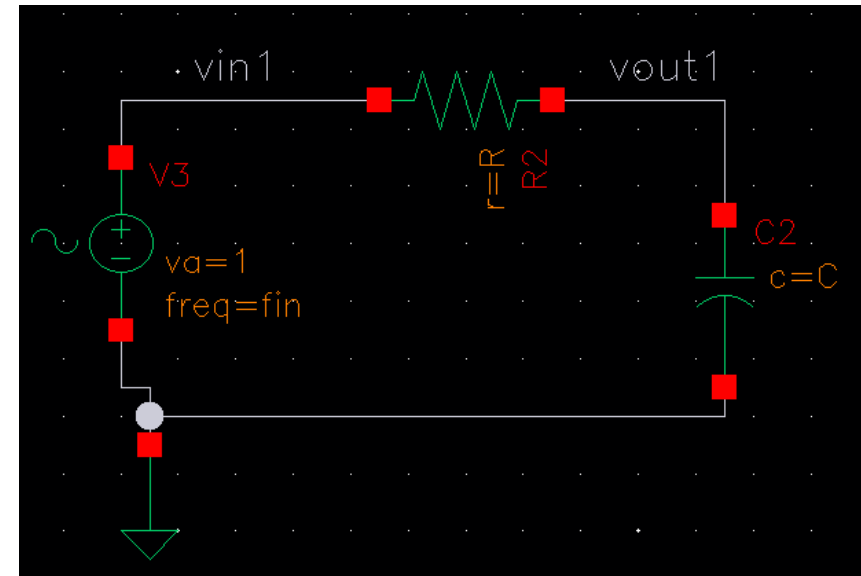
Set-up for fft

Library Name	analogLib
Cell Name	vsin
View Name	symbol
Instance Name	V3
Add Delete Modify	
User Property	Master Value Local Value
Ivsignore	TRUE
CDF Parameter Value	
First frequency name	
Second frequency name	
Noise file name	
Number of noise/freq pairs	0
DC voltage	
AC magnitude	1 V
AC phase	
XF magnitude	
PAC magnitude	
PAC phase	
Delay time	
Offset voltage	
Amplitude	1 V
Initial phase for Sinusoid	
Frequency	fin Hz
Amplitude 2	

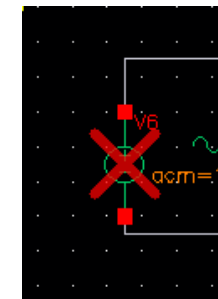
For ac simulation

For tran simulation

Previous RC net-work



To ignore source:
Right click → Ignore



Performing a FFT

To get a decent FFT some points are important to remember:

- FFT algorithms work most efficient when using a 2^N number of sample points. Although with current processors this is hardly an issue. For this example $N=16$,

NFFT (no. of sample points) = 65536.

- With a given sample rate f_s , the frequency spectrum has a frequency resolution of

$f_{\text{BIN}} = f_s / \text{NFFT}$. (Also called the resolution bandwidth-RBW).

- To get a decent FFT plot, and avoid signal skirts, it is necessary to ensure that the input frequency satisfies

$f_{\text{IN}} = (f_s / \text{NFFT}) * X_{\text{INT}}$, where X_{INT} is an integer. In other words, this is ensuring the input frequency *falls on one of the FFT bins*.

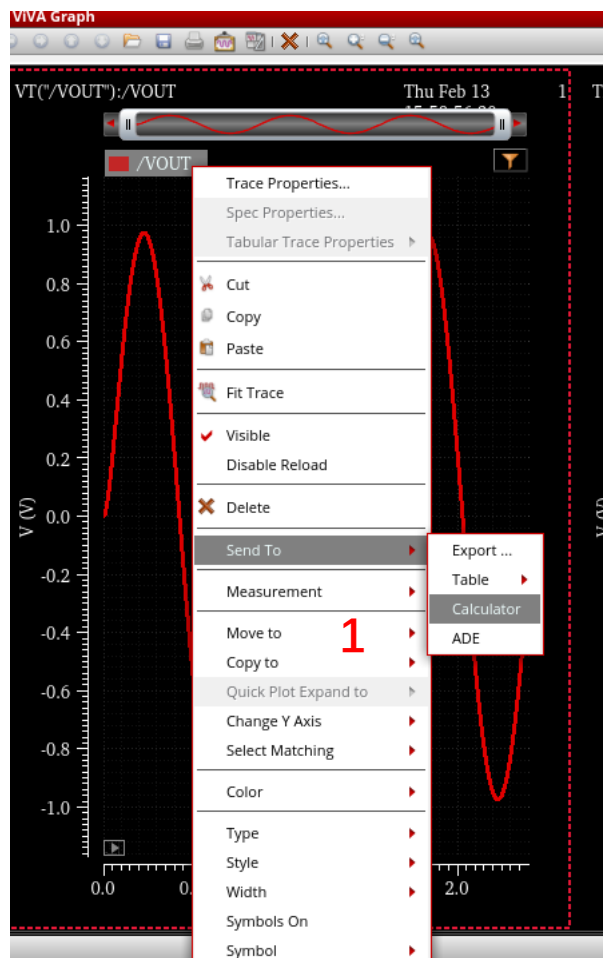
- For this example and f_{BIN} of 250Hz sets **$f_s = 250 * 2^{16} = 16.384\text{MHz}$**
- Total simulation time to get an FFT is **$t_{\text{sim}} = 1/f_{\text{BIN}} = 4\text{ms}$** . It is a good practice to give some additional time to allow for initial transient settling. In this case 200us is given (**$t_{\text{sim}} = 4.2\text{ms}$**)

The screenshot shows a 'Setup' window for a simulation. It has two main panes. The left pane is titled 'Filter' and contains a tree view with the following items: 'hw_test_RC_filter_1', 'Simulator spectre', and 'Analyses'. Under 'Analyses', there are two checked items: 'tran' with a value of '0 4.2m conservative' and 'ac' with a value of '1 1G Automatic Start-Stop'. Below these are links for 'Click to add analysis' and 'Design Variables'. Under 'Design Variables', there are several variables listed: 'C' (30n), 'R' (1k), 'fin' (Xint*fbin), 'fbin' (250), 'Xint' (5), 'fs' (fbin*N_fft), and 'N_fft' (2**16). There is also a link for 'Click to add variable'. The right pane is a table with columns 'Name', 'Type', and 'Value'. It contains three rows: the first row has 'Name' as an empty cell, 'Type' as 'signal', and 'Value' as '/VIN'; the second row has 'Name' as an empty cell, 'Type' as 'signal', and 'Value' as '/VOUT'; the third row has 'Name' as '/VOUT', 'Type' as 'expr', and 'Value' as 'VT("/VOUT")'.

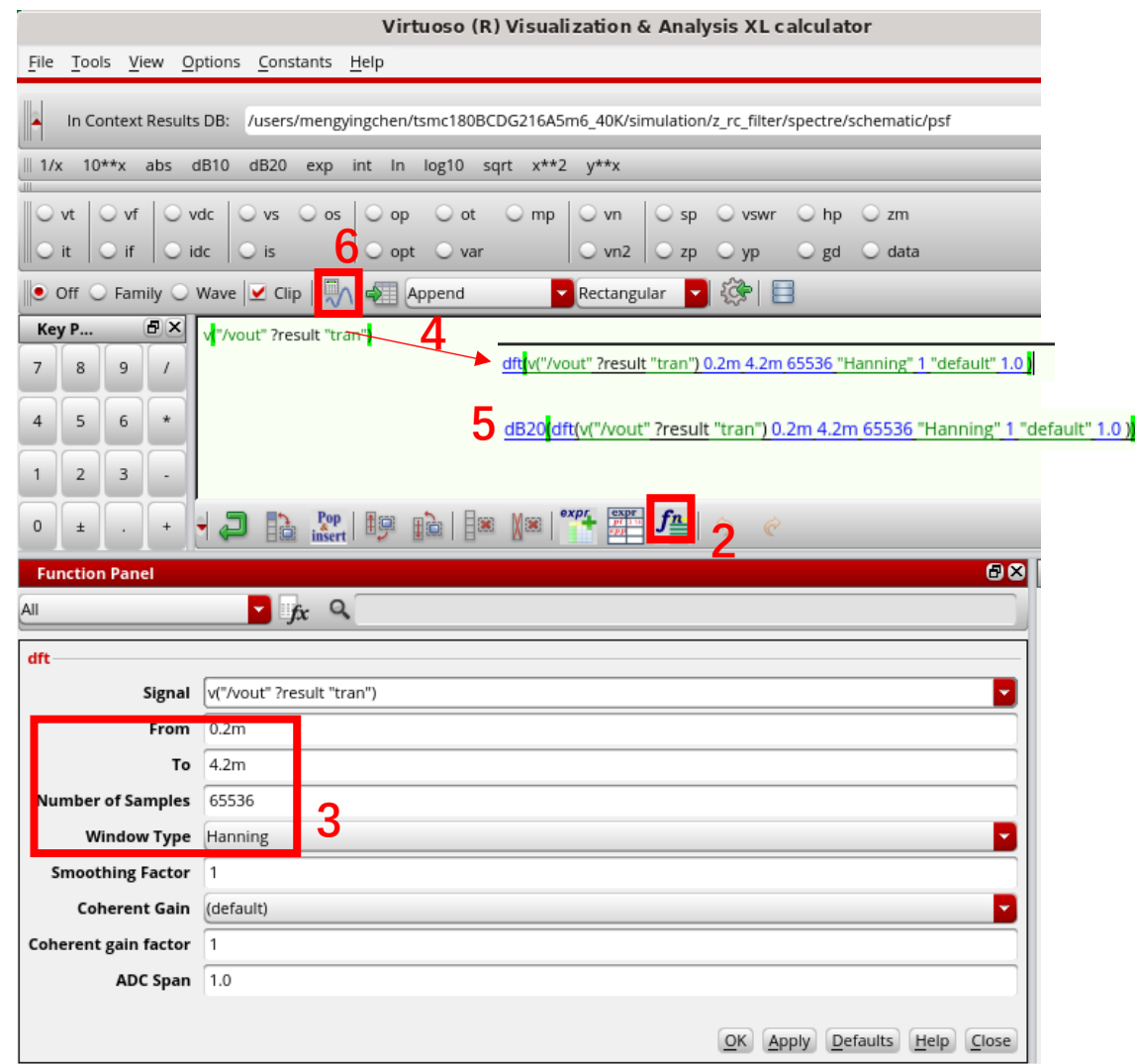
Name	Type	Value
	signal	/VIN
	signal	/VOUT
/VOUT	expr	VT("/VOUT")

FFTs and Signal-to-Noise-and Distortion (SINAD)

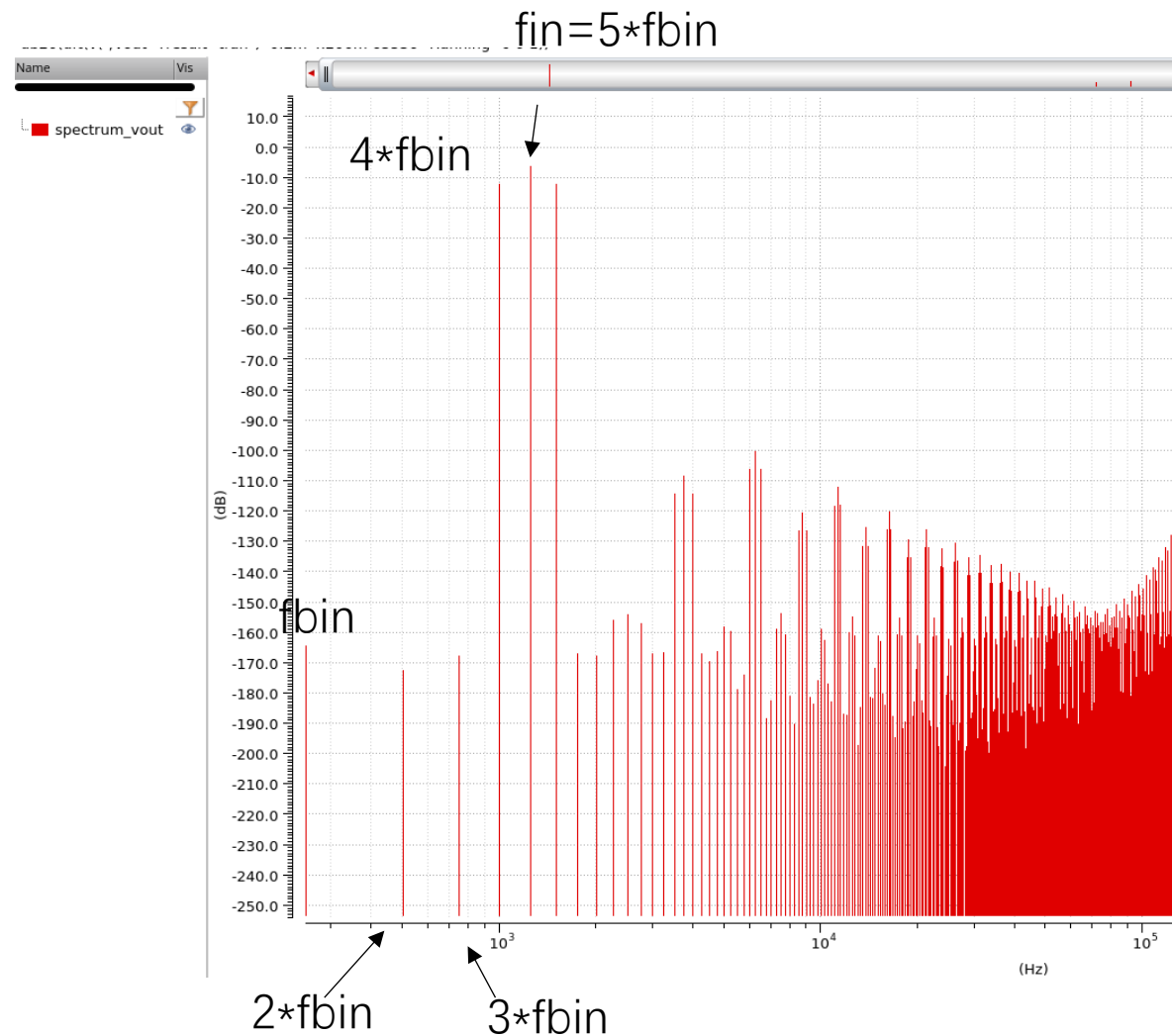
1. Right click waveform → send to → calculator
2. Select 'dft' in Function Panel



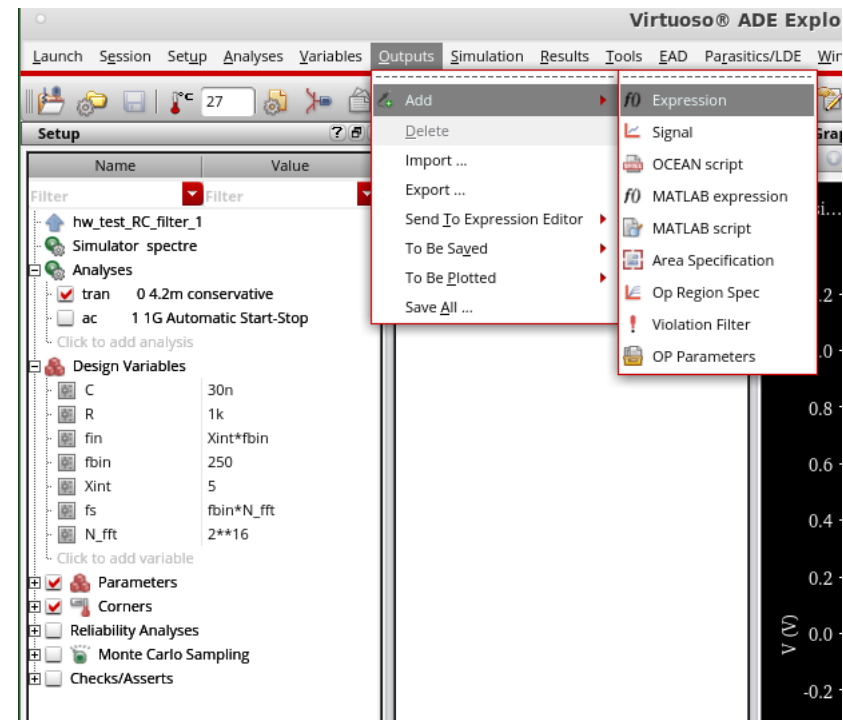
3. Enter dft setting as follow
4. Click apply get the equation
5. Type dB20(...)
6. Click 'plot'



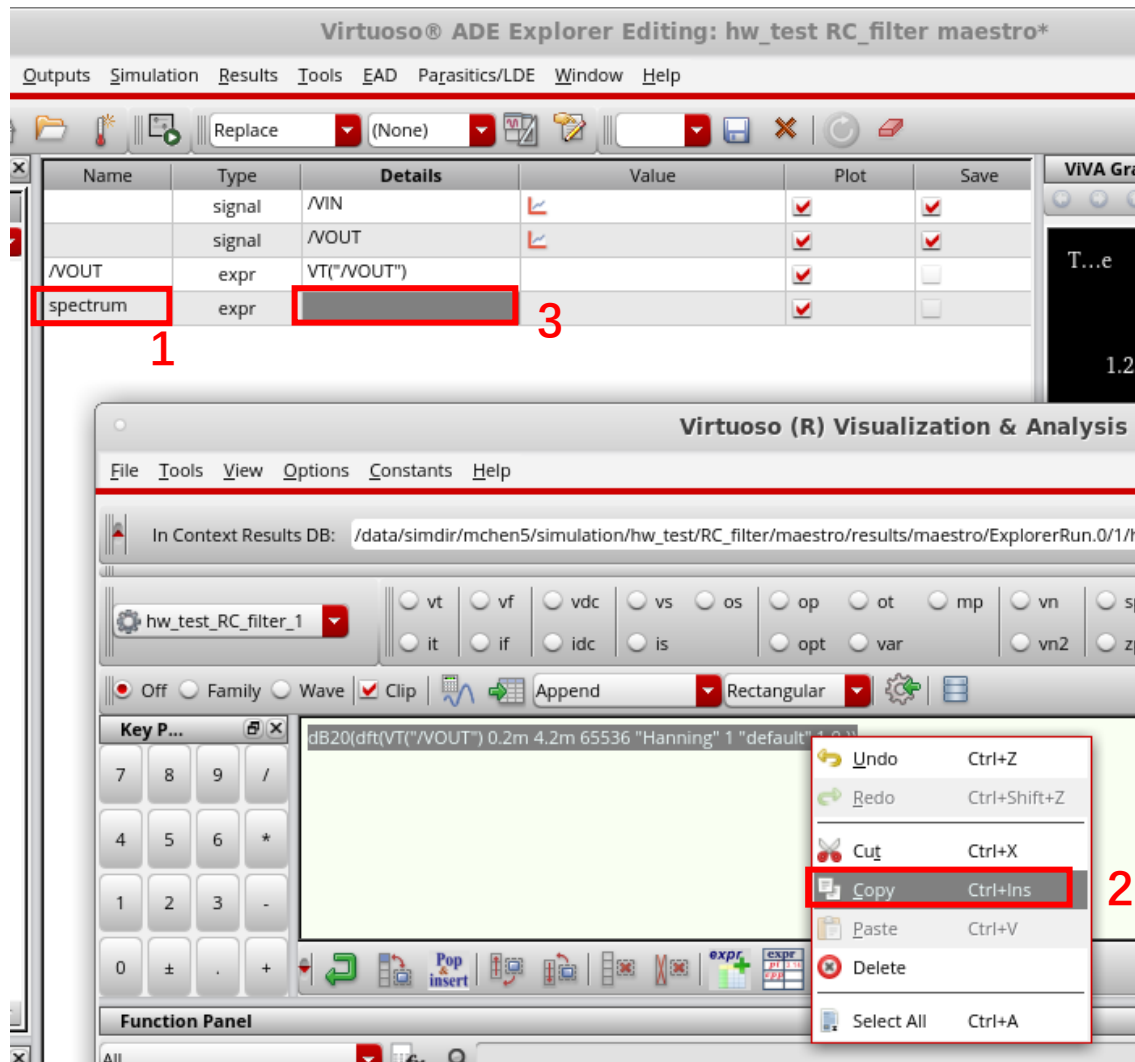
FFT spectrum



Outputs→Add→Expression



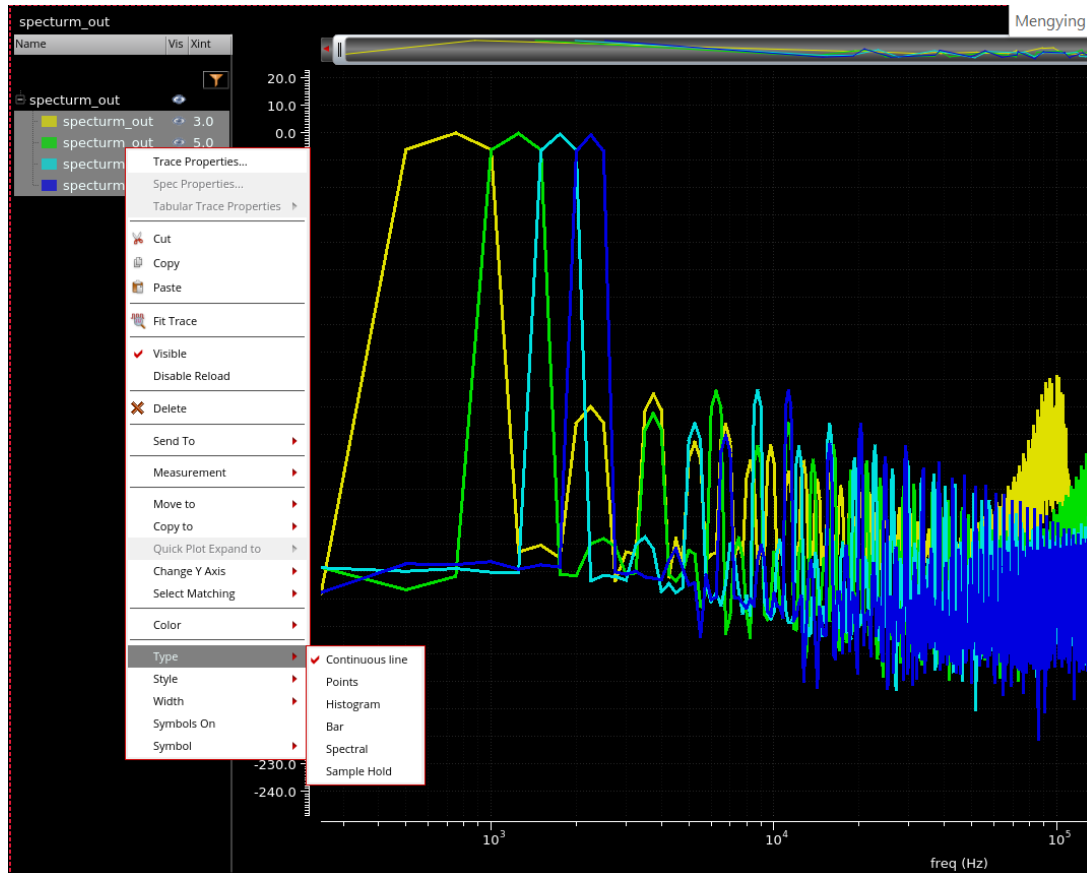
Save fft spectrum to output expression



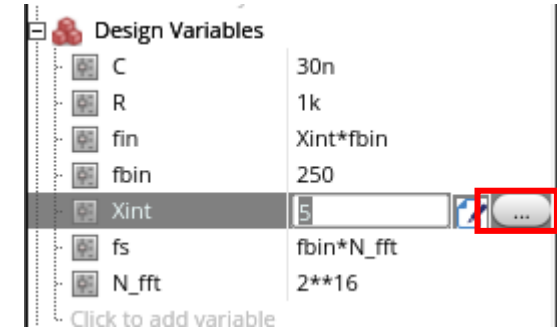
1. Type in 'spectrum' as expression **name**
2. Copy the expression from calculator
3. Paste the expression to **detail**

Sweep parameter

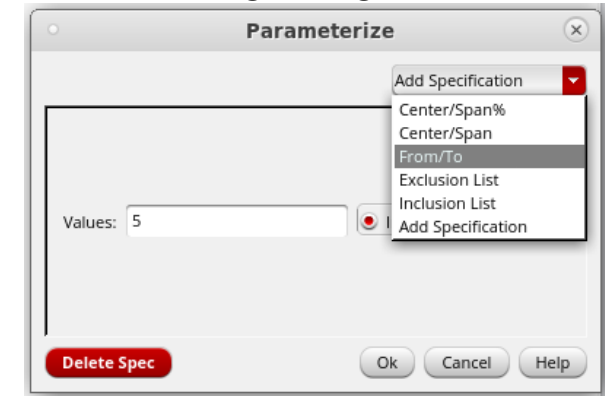
- To sweep the input frequency and plot the FFT at one go. a parametric sweep may be performed..



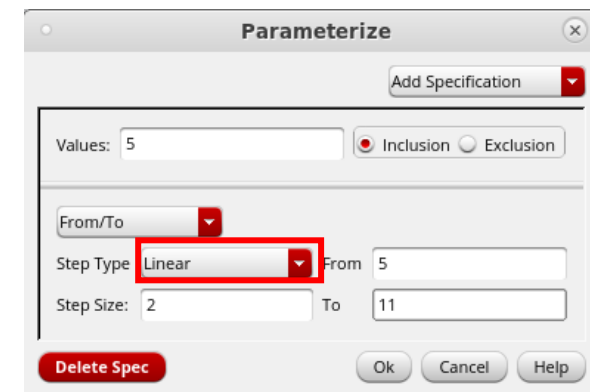
1. Double click **Xint**
2. Click ...



3. Select 'From/To'



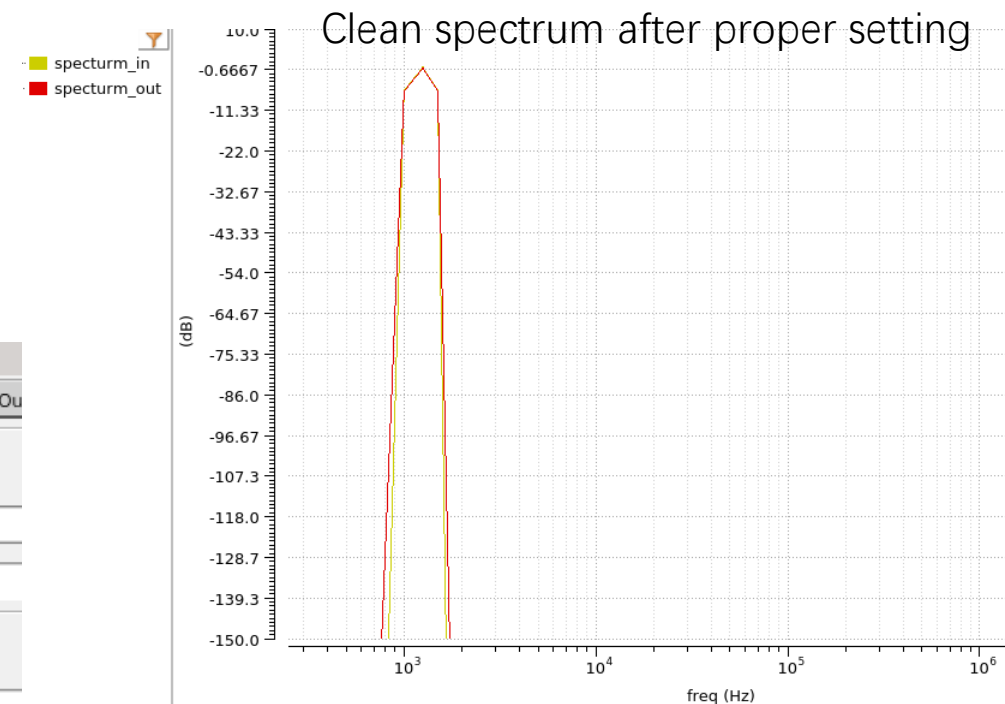
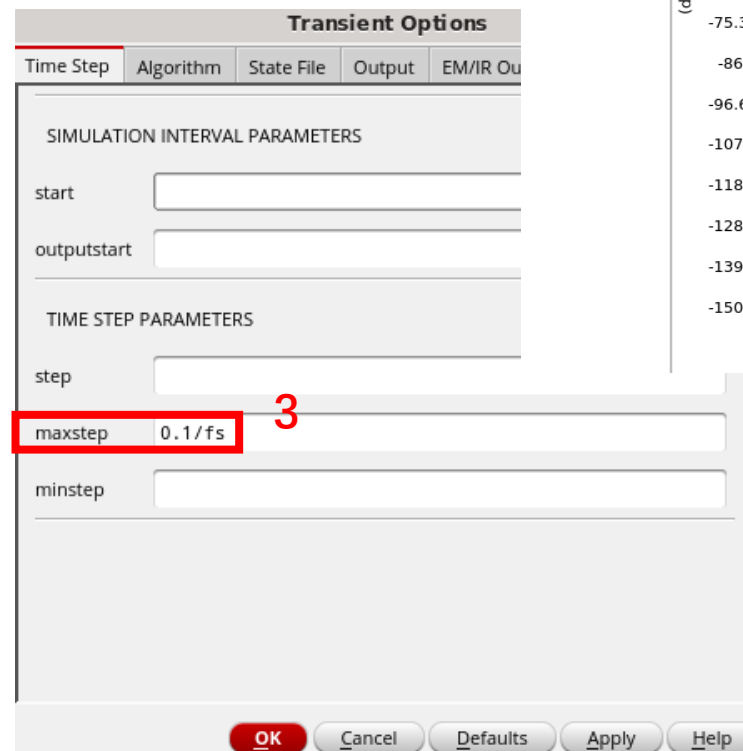
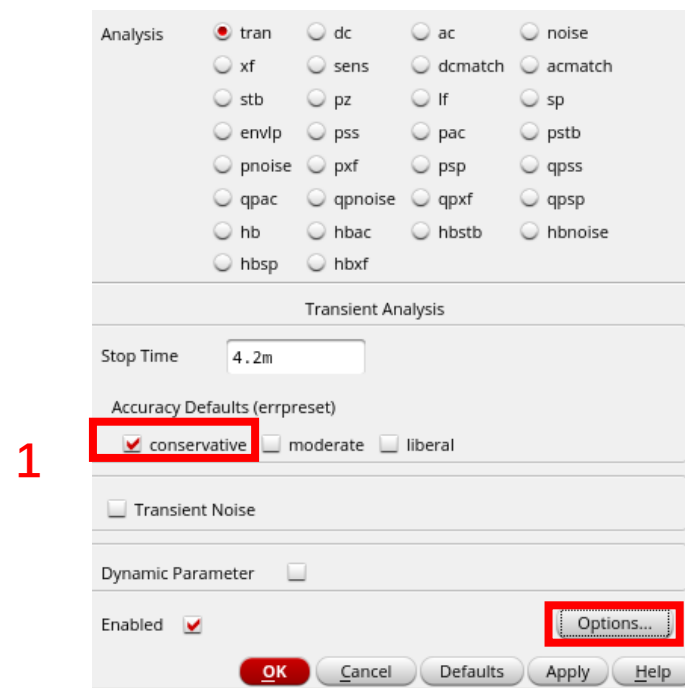
4. Configure as follows



Performing a FFT

Why a pure sin wave input has so many distortion?

- Setting in transient analysis



Exercise 5

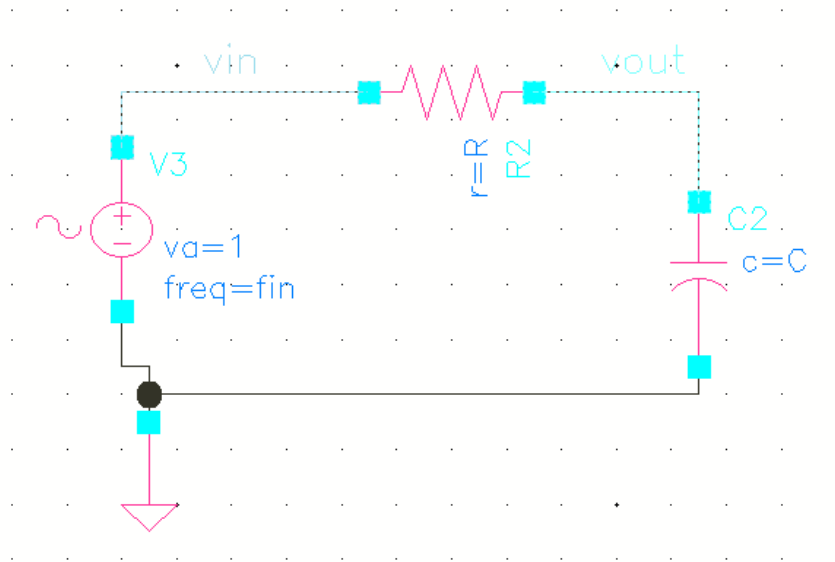
- FFT simulation set-up as follow

No. of sample points	$N_{fft}=2^{16}=65536$
Resolution bandwidth	$f_{BIN}=250\text{Hz}$
Input signal bin number	$X_{int}=5$
Input signal frequency	$f_{in}=X_{int}*f_{BIN}$
Sample frequency	$f_s=f_{BIN}*N_{fft}$
Simulation time	$1/f_{BIN}+200\mu=4.2\text{m}$

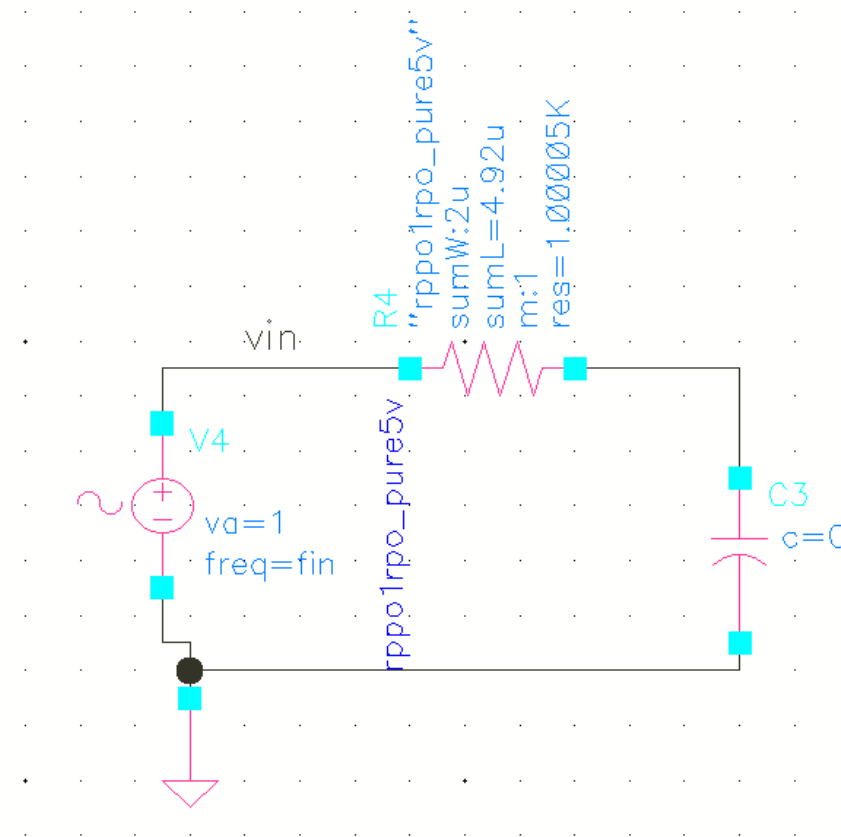
- Run tran simulation, plot waveform, select waveform, 'Measurements>spectrum'
- Save the spectrum expression into 'outputs'
- Set-up 'parametric analysis' for multiple simulation
- Add 'maxstep' restriction into tran simulation
-

Small testbench to see distortion

- RC filter with ideal R
- R: analogLib/res

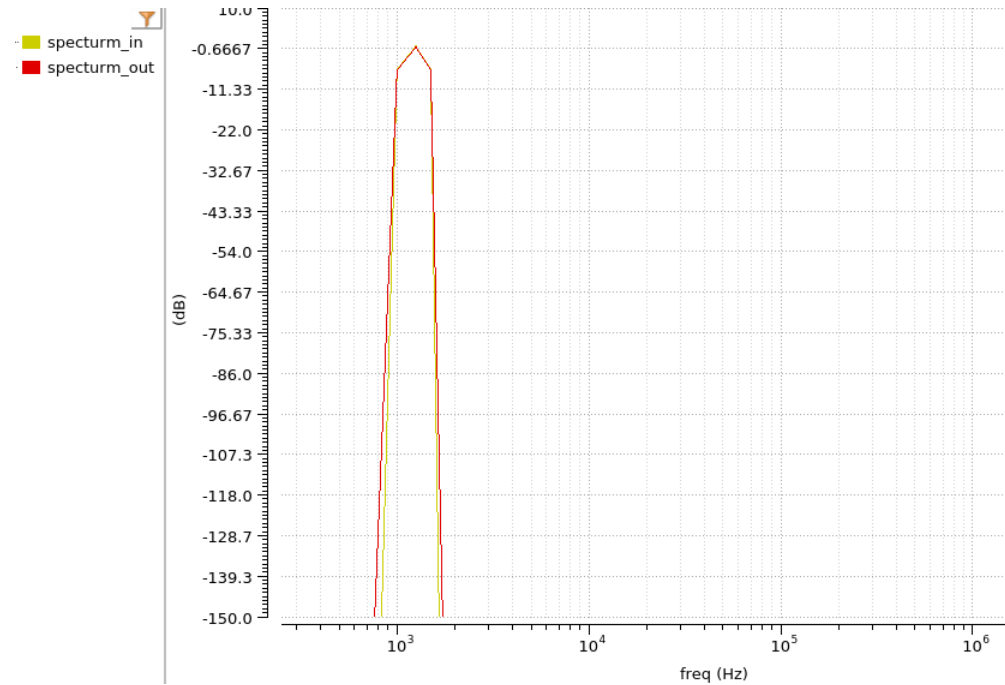


- RC filter with real R
- R: tsmc18/rppo1rpo_pure5v

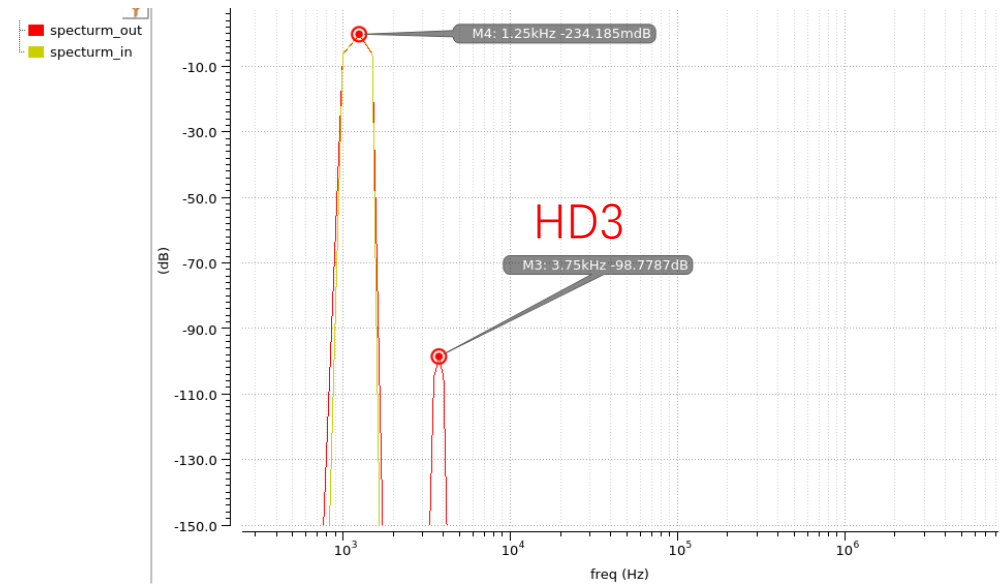


Small testbench to see distortion

- RC filter with ideal R

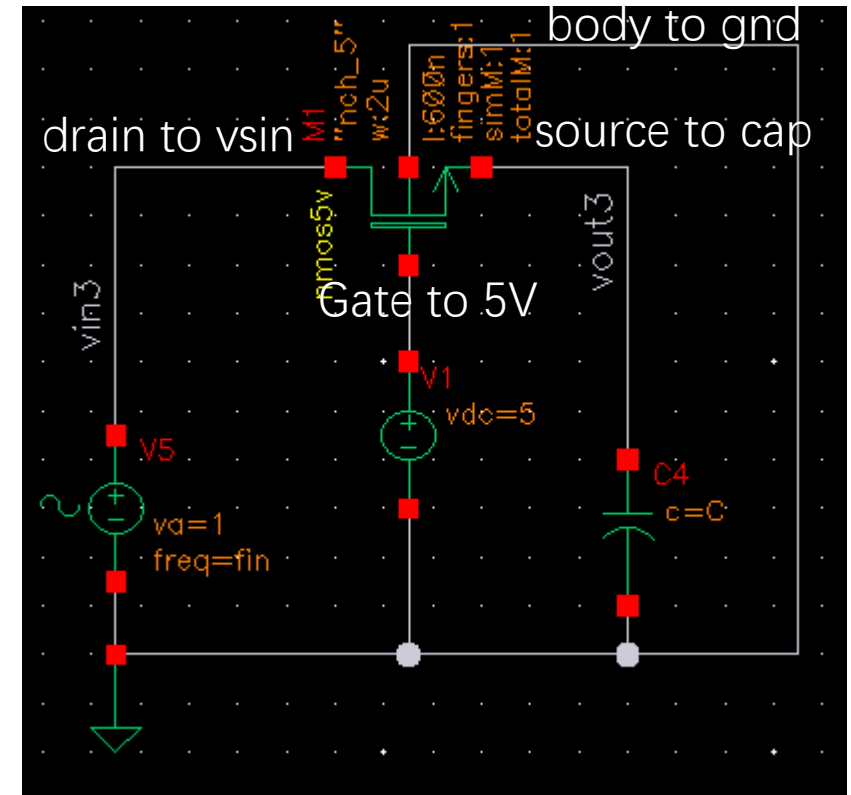
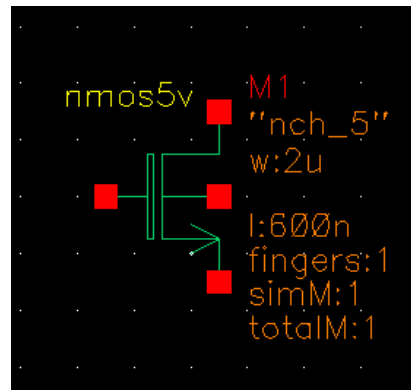
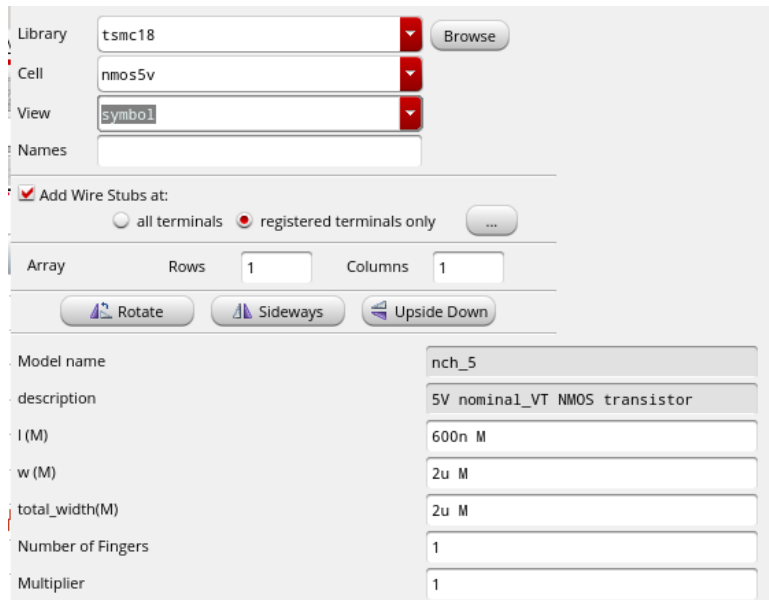


- RC filter with real R



Add Mosfets

- Create an instance using the shortcut “i” on the keyboard
- Select ‘tsmc18’ as library, ‘nmos5v’ as cell
- Rotate NMOS and connect it into circuit



Print nmos dc operation point

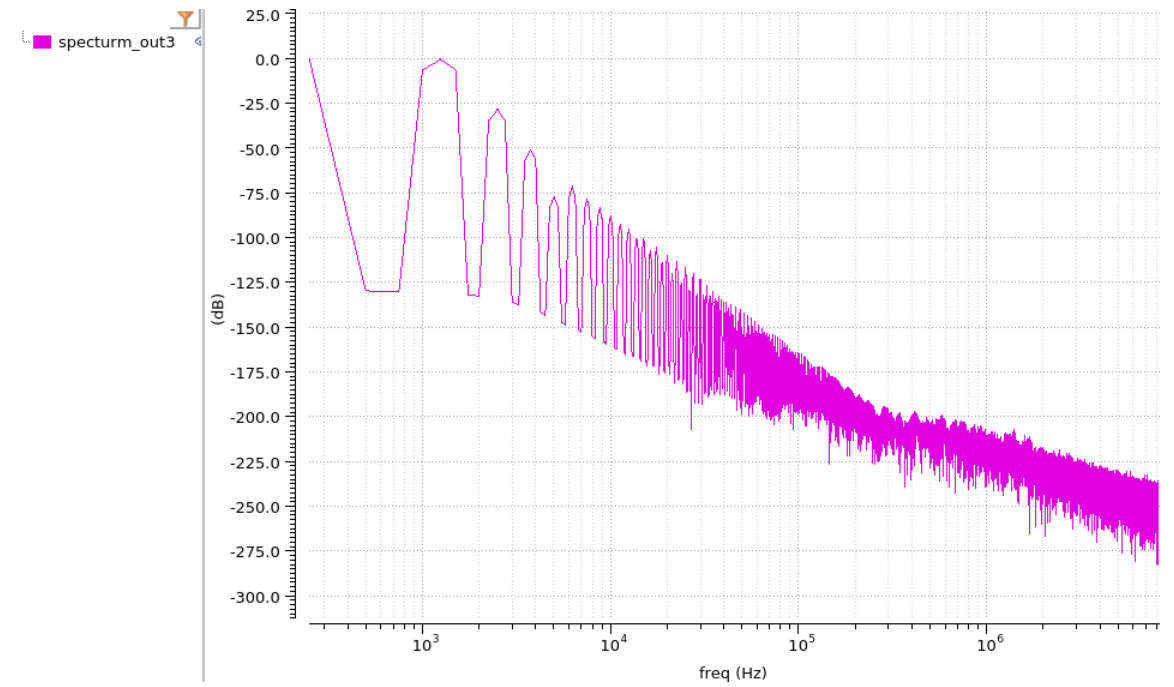
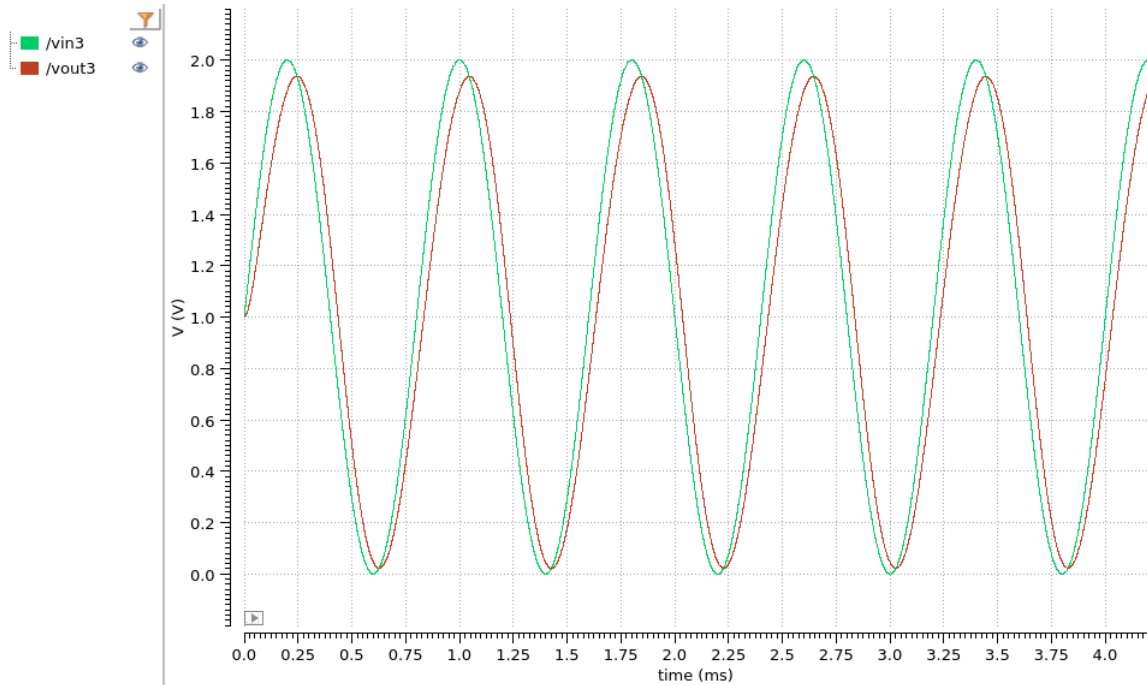
- Set vsin as offset voltage=1V, Amplitude=1V
- Run dc simulation
- ADE window: Results/Print/DC operation points
- Click on NMOS to select it to show in 'results display window'

Cell Name	vsin
View Name	symbol
Instance Name	V5
AddDeleteModify	
User Property	Master ValueLocal Value
lvignore	TRUE
CDF Parameter	Value
First frequency name	
Second frequency name	
Noise file name	
Number of noise/freq pairs	0
DC voltage	
AC magnitude	1 V
AC phase	
XF magnitude	
PAC magnitude	
PAC phase	
Delay time	
Offset voltage	1 V
Amplitude	1 V
Initial phase for Sinusoid	
Frequency	fin Hz

signal	OP("/M1" "??")
qgdovl	1.126f
qgi	10.9f
qgsovl	1.126f
qjd	-1.831f
qjs	-1.831f
qs	-4.19f
qsi	-4.19f
rdeff	982.8m
region	1
reversed	0
rgate	0
rgbd	0
ron	1.1K
rout	1.1K
rseff	982.8m
self_gain	243.8p
trise	0
ueff	37.81m
vbs	-1
vdb	1
vds	1.102n
vdsat	1.38
vdsat_marg	NaN
vdss	1.38
yearly	2.205n
vgb	5
vgd	4
vgs	4
vgt	2.838
vsat_marg	-1.38
vsb	1
vth	1.162
vth_drive	NaN

Small testbench to see distortion

- Run tran simulation
- Repeat fft expression



Optional task

- Replace ideal res with real res to see the distortion
- Add 'nmos5v' and build the testbench around it
- Run dc simulation and print dc operation point
- Run tran simulation and show fft spectrum

Reference

■ Brightspace tutorial1

Tutorial 1 ▼

Upload / Create ▼ Existing Activities ▼

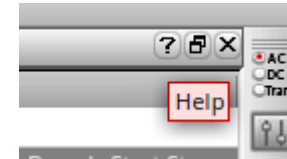
Tutorial-1	PDF document	✓
Tutorial-1 Recording	Video	✓
Server connecting	Word Document	✓

■ Brightspace cadence/spectre tutorial

[cadence/spectre](#) 2

[tutorial](#)

■ Cadence Help



Mini-project

■ Miniproject position

